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GSD CarPass Depot App

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

Rapid growth in campus vehicle ownership has made effective parking management a critical challenge for educational institutions. Manual pass verification at entrance and exit checkpoints causes severe traffic congestion during peak hours, imposing heavy workloads on security staff and causing delays for students and faculty. Beyond daily inconvenience, these operational bottlenecks hinder effective space utilization and compromise campus security [1], [2], [3], [4].

Furthermore, when parking-related issues arise, such as illegal parking or vehicles obstructing designated areas, guards and General Services Department (GSD) staff often face challenges in promptly contacting vehicle owners. The lack of an efficient communication and monitoring system can delay the resolution of parking violations and hinder the enforcement of campus parking regulations. As a result, the overall management, safety, and accessibility of campus parking facilities may be negatively affected, emphasizing the need for a more efficient GSD CarPass Depot App [1], [5], [6].

At the University of Iloilo, the General Services Department (GSD) plays a crucial role in maintaining the efficiency, safety, and functionality of campus operations. The department is responsible for a wide range of services, including building maintenance, campus safety and security, management of lost and found items, and the implementation of vehicle entry and parking policies. Through these responsibilities, the GSD helps ensure that the university environment remains organized, secure, and conducive to learning and daily operations.

One of the GSD's key responsibilities is managing vehicle entries and parking allocations within the university. Currently, the university relies on a traditional manual car pass system, requiring applicants to physically visit multiple offices and complete paper-based procedures to obtain

vehicle permits. While this process has been functional in the past, the growing number of vehicles entering the campus has highlighted several challenges, including longer processing times, increased administrative workload, and traffic congestion during peak hours due to manual sticker verification at the gates. In addition, the absence of a centralized digital repository for vehicle records makes it difficult to monitor real-time campus occupancy, verify authorized vehicles efficiently, and maintain accurate records for administrative purposes [1], [5], [6], [7], [8].

As institutions continue to adopt digital technologies to improve operational efficiency, reliance on physical stickers, paper-based records, and manual verification procedures has become increasingly ineffective in addressing the demands of modern campus management. These limitations can slow administrative processes and reduce the effectiveness of vehicle monitoring and parking enforcement [2], [3], [5], [7], [9].

Moreover, a well-managed vehicle access system is essential to the smooth daily operations of the university. The timely arrival and departure of faculty members, staff, and students are critical for maintaining academic schedules and administrative activities. Delays resulting from inefficient parking and vehicle verification procedures can negatively affect productivity and disrupt institutional operations. Therefore, adopting a technology-driven and centralized vehicle management system can significantly improve operational efficiency while providing a safer, more secure, and more convenient experience for all faculty members, staff, and students [2], [3], [4].

1.2 Statement of the Problem

The University of Iloilo's General Services Department (GSD) currently relies on a manual, paper-based GSD CarPass Depot App that has proven increasingly inadequate in addressing the demands of a growing campus population. The absence of a centralized and technology-driven solution has resulted in a range of operational inefficiencies that affect not only the administrative functions of

the GSD but also the daily experiences of faculty members, administrative staff, and students who own and use vehicles within the campus. As the volume of registered vehicles continues to rise, the limitations of the existing system have become more pronounced, manifesting in processing delays, security concerns, traffic congestion, and difficulties in enforcing campus parking regulations. The persistence of these challenges underscores the critical need for a modernized GSD CarPass Depot App that can effectively respond to the evolving needs of the university community. Specifically, this study seeks to address the following problems:

Inefficient Car Pass Application Process — The existing system requires applicants to physically visit multiple offices and complete paper-based procedures, resulting in longer processing times and a heavier administrative workload for GSD personnel.

Traffic Congestion at Entry and Exit Points — Manual sticker verification during peak hours causes significant traffic buildup at campus gates, leading to extended waiting times for vehicle owners and increased burden on security personnel.

Lack of a Centralized Digital Vehicle Repository — The absence of a unified digital system makes it difficult to monitor real-time campus parking occupancy, efficiently verify authorized vehicles, and maintain accurate and up-to-date vehicle records for administrative use.

Inadequate Enforcement of Campus Parking Regulations — Without a technology-driven monitoring system, enforcing parking policies and addressing violations in a timely and systematic manner remains a persistent challenge for the GSD.

Disruption to Academic and Administrative Operations — Delays caused by inefficient vehicle verification and parking management negatively affect the punctuality and productivity of faculty, staff, and students, ultimately disrupting the smooth daily operations of the university.

1.3 Objectives of the Study

1.3.1 General Objective

This study aims to design, develop, and implement a centralized, technology-driven and Parking Management System for the University of Iloilo's General Services Department (GSD) that will comprehensively address the growing inefficiencies and operational challenges associated with the current manual, paper-based approach to vehicle access control and parking administration. Recognizing the critical role that an efficient vehicle management system plays in sustaining the smooth and uninterrupted flow of daily academic and administrative operations, the study seeks to harness the capabilities of modern digital technology to transform and modernize the way the university handles vehicle entry, parking allocation, permit processing, and regulation enforcement within its campus premises. By transitioning to a fully digital system, the study intends to streamline the car pass application and approval process, establish a centralized digital repository for all vehicle-related records, and introduce a faster and more reliable vehicle verification mechanism that minimizes traffic congestion at campus entry and exit points during peak hours, thereby reducing the workload of security personnel and improving the overall flow of vehicle traffic within and around the university.

1.3.2 Specific Objectives

Specifically, the study aims to:

Develop a QR Code-Based Digital Car Pass Generation System — Design and implement a digital car pass system that allows GSD staff to generate, manage, and issue QR code-based vehicle permits containing key vehicle information such as the plate number, owner name, owner email, and validity period, effectively replacing the existing manual, paper-based sticker system.

Implement an Automated QR Code Distribution via Email — Enable the system to automatically deliver generated QR codes directly to vehicle owners through email, eliminating the need for applicants to physically visit multiple offices to claim their vehicle permits and reducing the administrative workload of GSD personnel.

Design a QR Code Scanning and Verification Mechanism for Security Guards — Develop a scanning interface that allows security guards to quickly verify the validity of vehicle QR codes at campus entry and exit points, automatically detecting the status of each QR code as accepted, expired, or revoked, and logging each scan in real time to minimize manual checking and reduce traffic congestion during peak hours.

Establish a Centralized Digital Repository for Vehicle Records and Entry Logs — Create a unified digital database that stores all QR code records, vehicle information, and entry and exit history, enabling authorized personnel to efficiently retrieve, monitor, and manage vehicle data and campus parking occupancy at any given time.

Provide an Administrative Dashboard for Real-Time Parking Monitoring and Reporting — Develop an admin dashboard that displays real-time data on parking slot availability, total vehicle entries and exits, active QR codes, and recent activity logs, enabling the GSD administration to make informed decisions regarding parking management and space utilization.

Enable Role-Based Access Control for System Users — Implement a secure, role-based access control mechanism that assigns distinct permissions and functionalities to three user roles — Admin, Staff, and Guard — ensuring that each user can only access the features and data relevant to their designated responsibilities within the system.

1.4 Scope and Delimitation

Scope

This study covers the development and implementation of the GSD CarPass Depot App, a web-based application designed exclusively for the General Services Department (GSD) of the University of Iloilo. The system aims to modernize and streamline the department's vehicle access control and parking management operations through the use of QR code-based vehicle permits and digital record management.

The system accommodates three user roles with distinct functionalities. The Admin manages system user accounts, including the creation, viewing, and deletion of Admin, Staff, and Guard accounts; configures and adjusts total parking slot capacity; monitors real-time parking occupancy; views vehicle entry and exit activity logs; and generates reports through a comprehensive administrative dashboard. The Staff generates and issues QR code-based vehicle permits, manages vehicle owner information such as plate numbers and email addresses, delivers QR codes to vehicle owners via automated email, renews and reactivates expired QR code permits, and views the complete history of all generated QR codes. The Guard scans and verifies QR codes at campus entry and exit points using a dedicated scanning interface, logs each scan transaction with its corresponding status (accepted, expired, or revoked), and views personal scan history.

The GSD CarPass Depot App covers the generation and issuance of QR code-based vehicle permits containing essential information, including the vehicle plate number, owner name, owner email address, and permit validity period. An automated email delivery mechanism transmits the approved QR code-based vehicle permit directly to the registered email address of the vehicle owner upon generation by GSD staff. The system also supports the renewal and reactivation of expired QR code permits, allowing GSD staff to extend the validity of existing vehicle passes without generating entirely new QR codes.

The GSD CarPass Depot App provides real-time monitoring of parking slot availability and occupancy, automatically updating parking status based on QR code scan data recorded at campus

entry and exit points. A centralized digital repository maintains all vehicle records, QR code permits, ownership information, and vehicle entry and exit history, accessible only to authorized GSD personnel. Activity logs and reports documenting vehicle entry and exit transactions, QR code scan results, and overall parking utilization are generated by the system and are accessible to the Admin for administrative and reporting purposes. Profile management is also supported, allowing system users to update their account email address and password credentials.

The GSD CarPass Depot App is built using the Flask Python web framework for the backend, with an HTML, CSS, and JavaScript frontend, and a MySQL relational database for data storage and management. It does not have a dedicated mobile application; however, the QR code scanning interface used by security guards may be accessed through a mobile web browser, as the scanning feature utilizes the device's camera for QR code reading. The system requires a stable internet connection for all core functionalities, including QR code generation, email delivery, scanning and verification, and real-time parking monitoring. A secure authentication mechanism restricts access to authorized users only, with role-based permissions ensuring that each user can access only the features relevant to their designated role. The GSD CarPass Depot App is designed exclusively for deployment and use within the operational context of the University of Iloilo's General Services Department and is not intended for use by vehicle owners or the general public. Delimitation

The study is bounded by the following limitations in terms of the scope and coverage of the GSD CarPass Depot App.

The system does not cover the online submission of car pass applications by vehicle owners. Applicants are still required to manually accomplish application forms, pay the corresponding car pass fees, and personally submit their documentary requirements to the GSD office prior to QR code permit generation. Accordingly, the system does not include an automated payment

processing feature or online collection of car pass fees, as the collection, verification, and recording of fees remain a manual responsibility of the GSD and are outside the scope of the system.

The system does not incorporate physical detection of vehicles using sensors, cameras, or any hardware-based vehicle detection technology for monitoring actual parking space utilization within the campus premises. Parking occupancy tracking is based solely on QR code scan data recorded at campus entry and exit points. As such, the system does not automatically detect, flag, or report parking violations, unauthorized parking, or vehicles occupying restricted areas. Physical monitoring and enforcement by security personnel remain necessary for these concerns. Real-time GPS tracking or location monitoring of vehicles within the campus premises is likewise not supported.

The system does not include a notification or alert system directed to vehicle owners regarding the status of their QR code permits, such as upcoming expiration reminders, renewal notices, or parking violation alerts. The system also does not support integration with third-party systems such as the university's enrollment system, human resource system, or any other existing institutional database. Furthermore, the system does not cover the generation of printed physical car pass stickers or any other form of physical vehicle permit, nor does it cover the management of pedestrian access, visitor passes, or non-motorized vehicles such as bicycles within the campus premises.

The system is accessible only via a web browser and does not include a dedicated mobile application for any of the three user roles on Android or iOS platforms. All core functionalities of the system, including QR code generation, email delivery, scanning and verification, and real-time parking monitoring, require a stable internet connection, as offline or local network operation is not supported. Finally, the GSD CarPass Depot App is designed exclusively for the operational requirements of the University of Iloilo's General Services Department and is not intended for

deployment in other institutions without significant modifications to accommodate different parking management structures and policies.

1.5 Significance of the Study

This study holds considerable significance as it addresses the real and pressing operational challenges faced by the University of Iloilo's General Services Department in managing vehicle access and parking within the campus. The development and implementation of the GSD CarPass Depot App is expected to bring meaningful improvements to the operational efficiency, security, and convenience of campus vehicle management. The following are the primary beneficiaries of this study:

General Services Department (GSD). The GSD stands as the primary beneficiary of this study. The system significantly reduces the administrative burden associated with the manual processing, issuance, and distribution of vehicle permits by digitizing the generation and delivery of QR code-based car passes. With a centralized digital repository for vehicle records and a real-time dashboard for monitoring parking occupancy and activity logs, the GSD will be better equipped to manage campus vehicle access efficiently, enforce parking policies, and maintain accurate and organized records without relying on paper-based procedures.

GSD Staff. The system directly improves the day-to-day workflow of GSD staff members by providing a streamlined digital interface for generating, issuing, renewing, and managing QR code-based vehicle permits. The automated email delivery of approved car passes eliminates the need for staff to manually distribute physical permits, allowing them to dedicate more time and attention to other essential administrative responsibilities within the department.

Security Guards. Security personnel stationed at campus entry and exit points will benefit from a faster and more reliable vehicle verification mechanism. The QR code scanning interface enables guards to instantly determine the validity status of a vehicle permit as accepted, expired, or

revoked, minimizing the time spent on manual sticker verification and reducing the occurrence of traffic congestion during peak hours. This also lessens the physical workload of guards and enhances the overall security of campus entry and exit operations.

Faculty Members and Administrative Staff. Faculty and administrative staff who own and use vehicles within the campus will experience a more convenient and efficient car pass issuance process. The automated delivery of QR code permits directly to their registered email addresses eliminates the need to make multiple visits to the GSD solely to claim their vehicle permits, saving time and effort while ensuring timely receipt of their approved passes. Additionally, the QR code renewal feature allows vehicle owners with expired permits to have their passes reactivated more efficiently without undergoing the full application process again.

University Administration. The University of Iloilo's administration will benefit from the improved organizational efficiency and enhanced security that the system brings to campus operations. The availability of real-time data on parking occupancy, vehicle entries and exits, and permit activity provides administrators with valuable insights for informed decision-making regarding parking infrastructure, policy enforcement, and resource allocation. The system also supports the university's broader institutional goal of adopting modern digital technologies to improve operational efficiency, enhance campus safety and security, and provide a better environment for all members of the university community.

Future Researchers and Developers. This study serves as a foundational reference and practical resource for future researchers and developers who may seek to further enhance, expand, or build upon the GSD CarPass Depot App. The system's modular architecture, role-based access control design, and documented codebase provide a solid and scalable foundation for future development efforts. Future developers may extend the system by incorporating additional features such as online car pass application and submission, automated payment processing, hardware-based

vehicle detection, mobile application development, or integration with other existing university systems. The insights, methodologies, and lessons documented in this study can likewise guide future researchers in addressing similar challenges in other institutional or organizational contexts, contributing to the broader body of knowledge in the field of campus management systems and information technology development.

1.6 Definition of Terms

The following terms are defined both conceptually and operationally to provide clarity and a common understanding of the key concepts used throughout this study:

Vehicle Access and Parking Management System. Conceptual — A vehicle access and parking management system refers to an integrated set of tools, technologies, and procedures designed to regulate, monitor, and control the entry, exit, and parking of vehicles within a defined area or facility, with the goal of improving security, efficiency, and space utilization [10]. Operational — In this study, the Vehicle Access and Parking Management System refers to the web-based application developed for the General Services Department of the University of Iloilo that manages the generation and issuance of QR code-based vehicle permits, monitors real-time parking occupancy, and records vehicle entry and exit transactions within the campus premises.

QR Code (Quick Response Code). Conceptual — A QR code is a two-dimensional matrix barcode that stores information in a pattern of black and white squares and can be scanned using a digital device such as a smartphone or barcode reader to quickly retrieve encoded data [11]. Operational — In this study, QR code refers to the unique, digitally generated code embedded in each vehicle permit issued by the GSD staff through the system, which contains the vehicle owner's information and permit validity details and is scanned by security guards at campus entry and exit points for verification purposes.

Car Pass. Conceptual — A car pass is an official authorization document or permit issued by a governing authority or institution that grants a vehicle owner the right to access, enter, and park within a designated area or facility [12]. Operational — In this study, a car pass refers to the QR code-based digital vehicle permit generated and issued by GSD staff through the system to authorized faculty members, administrative staff, and students of the University of Iloilo, granting them the right to enter and park within the campus premises.

General Services Department (GSD). Conceptual — A general services department is an organizational unit within an institution responsible for managing and overseeing a wide range of support and administrative services, including facilities maintenance, security, and the implementation of institutional policies and procedures [13]. Operational — In this study, the General Services Department (GSD) refers to the specific department of the University of Iloilo responsible for managing vehicle access, parking allocation, and the issuance of car pass permits within the campus, and which serves as the primary end-user and beneficiary of the developed system.

Real-Time Monitoring. Conceptual — Real-time monitoring refers to the continuous and immediate observation, tracking, and recording of data or activities as they occur, enabling users to access up-to-date information and respond promptly to changing conditions [14]. Operational — In this study, real-time monitoring refers to the system's capability to instantly update and display the current parking occupancy, vehicle entry and exit counts, and QR code scan activity on the administrative dashboard as transactions are recorded at campus entry and exit points.

Role-Based Access Control (RBAC). Conceptual — Role-based access control is a security mechanism that restricts system access and functionalities to users based on their assigned roles within an organization, ensuring that individuals can only perform actions and view information relevant to their designated responsibilities [15]. Operational — In this study, role-based access

control refers to the system's security feature that assigns distinct permissions and functionalities to three user roles — Admin, Staff, and Guard — preventing unauthorized access to features and data outside the scope of each user's designated role within the GSD.

Web-Based Application. Conceptual — A web-based application is a software program that is hosted on a remote server and accessed by users through a web browser over the internet or a local network, without requiring installation on the user's device [16]. Operational — In this study, web-based application refers to the GSD CarPass Depot App developed for the General Services Department of the University of Iloilo, which is accessible through a web browser by authorized Admin, Staff, and Guard users to perform their respective vehicle management functions.

Parking Occupancy. Conceptual — Parking occupancy refers to the proportion or number of parking spaces within a facility that are currently in use at a given point in time, often used as a measure of parking demand and space utilization [17]. Operational — In this study, parking occupancy refers to the number of vehicles currently present within the campus parking area as tracked and displayed by the system's administrative dashboard, based on the QR code scan data recorded at campus entry and exit points.

Automated Email Delivery. Conceptual — Automated email delivery refers to the process by which a software system automatically generates and sends electronic mail messages to designated recipients upon the occurrence of a specific trigger or event, without requiring manual intervention from the sender [18]. Operational — In this study, automated email delivery refers to the system's feature that automatically transmits the approved QR code-based vehicle permit to the registered email address of the vehicle owner immediately upon its generation by GSD staff, eliminating the need for manual distribution of physical car passes.

QR Code Renewal. Conceptual — QR code renewal refers to the process of extending or updating the validity period of an existing QR code, allowing it to remain functional and authorized

beyond its original expiration date [19]. Operational — In this study, QR code renewal refers to the system feature that enables GSD staff to update the expiry date of an existing vehicle permit QR code, reactivating expired car passes and allowing vehicle owners to continue using their previously issued QR codes without the need for a completely new permit to be generated.

Entry and Exit Logs. Conceptual — Entry and exit logs are systematic records that document the movement of individuals or vehicles into and out of a facility, typically including details such as the time, date, and identity of the person or vehicle involved [20]. Operational — In this study, entry and exit logs refer to the digital records automatically generated and stored by the system each time a vehicle's QR code is scanned at a campus gate by a security guard, capturing the scan timestamp, QR code details, vehicle plate number, and whether the transaction was an entry or exit.

Centralized Digital Repository. Conceptual — A centralized digital repository is a single, unified storage system that consolidates and organizes digital data and records from multiple sources, making them accessible to authorized users from one central location [21]. Operational — In this study, centralized digital repository refers to the system's unified database that stores and maintains all vehicle permit records, QR code information, vehicle owner details, and entry and exit transaction logs, which can be accessed and managed by authorized GSD personnel through the web-based system.

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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

In an international context, Sravanthi and Rajender [22] investigated an intelligent QR code-based parking system integrated with feedback analytics among urban vehicle drivers, facility attendants, and smart city transportation administrators in an urban mobility setting. The researchers conducted their study to address the severe urban challenges of rising vehicle density, prolonged search times for vacant parking spaces, entry gate congestion, and the lack of actionable data for facility managers to optimize space utilization.

The study utilized an applied software development and experimental quantitative evaluation method to design, build, and test a web-enabled smart parking architecture that leverages QR codes for fast vehicle authentication alongside an analytics engine to collect real-time occupancy metrics and user feedback. Their findings revealed that transitioning from traditional manual paper ticketing to a scan-based QR code verification model significantly accelerated entry and exit validation times, reduced gate queue lengths during peak traffic hours, and provided facility managers with crucial operational insights for dynamic space allocation and service improvement. This aligns with the current study's premise that replacing paper-based procedures and manual physical sticker checks with a centralized, QR code-based digital vehicle management system

substantially improves administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points.

While Sravanthi and Rajender [22] focused on broad public urban mobility spaces and prioritized user feedback analytics to optimize commercial parking allocations, the GSD CarPass Depot App adapts a similar core QR code scanning architecture specifically tailored for an academic institutional environment—namely, the University of Iloilo's General Services Department (GSD). Both studies reinforce the core principle that digital, scan-based verification eliminates operational bottlenecks at access points. However, the current study diverges by incorporating a centralized web application built with Flask Python, automated email permit delivery directly to vehicle owners, permit renewal and reactivation features, and role-based access control (RBAC) specifically structured for Admins, Staff, and Security Guards to address the operational inefficiencies, administrative delays, gate congestion, and lack of real-time parking monitoring facing the General Services Department (GSD) at the University of Iloilo.

In an international context, Dokania et al. [7] investigated a QR code–based smart parking system among urban vehicle drivers, commercial parking lot vendors, and facility administrators in India. The authors conducted the study to resolve the persistent urban issues associated with traditional parking operations—specifically, the high capital investment and maintenance costs of sensor-heavy infrastructure, time loss caused by manual record-keeping and inefficient payment methods, difficulty in locating parked vehicles in high-capacity lots, and revenue loss for vendors due to unmonitored space utilization. The study utilized an experimental engineering and system development research design, building a cross-platform mobile application (using Flutter and Firebase ML Vision) that enables drivers to view available slots, complete digital payments, and receive a unique QR code serving as an e-ticket for automated entry, exit, and location tracking. Their findings revealed that substituting manual paper ticketing and costly hardware sensors with

a QR code-driven software architecture drastically reduced entry/exit processing times, eliminated administrative overhead, simplified payment verification, and provided vendors with real-time analytics on parking space allotment and revenue.

This aligns with the current study's premise that replacing paper-based procedures and manual physical sticker checks with a centralized, QR code-based digital vehicle management system substantially improves administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Dokania et al. [7] focused on commercial parking facilities, user-facing booking/payment mobile interfaces, and individual slot reservation for public motorists, the GSD CarPass Depot App adapts a similar core QR code verification framework specifically for an academic institutional environment—namely, the University of Iloilo's General Services Department (GSD). Both studies reinforce the core principle that digital scan-based verification eliminates operational bottlenecks at access points while centralizing vehicle records. However, the current study diverges by incorporating a centralized web application built with Flask Python, automated email permit delivery directly to vehicle owners, permit renewal/reactivation features, and role-based access control (RBAC) specifically designed for Admins, Staff, and Security Guards to address the administrative delays, paper-based workflows, and gate congestion facing campus operations.

In an international context, Omkar et al. [5] investigated a smart QR-based vehicle access control system featuring user-defined time selection and ID verification among vehicle drivers, gate security attendants, and facility administrators in an institutional parking setting. The authors conducted this study to address the persistent operational vulnerabilities of traditional vehicle gate management—specifically physical pass forgery, unauthorized vehicle entry, credential sharing, and the excessive time delays caused by manual paper logging at security checkpoints. To solve these problems, the study utilized an experimental system development and functional testing

research design to engineer and evaluate a contactless access framework that generates temporary, time-bound QR code passes containing encrypted driver ID credentials and custom valid-time durations. Their findings revealed that implementing digital QR code verification paired with automated time-validity enforcement significantly reduced gate check-in delays, eliminated pass duplication and credential sharing, and provided reliable, real-time access logging for security personnel.

This aligns with the current study's premise that replacing paper-based procedures, manual physical sticker checks, and physical office visits with an automated QR code-based verification system substantially enhances operational efficiency, security, and traffic flow within campus facilities.

While Omkar et al. [5] focused on user-configured temporary access durations and custom time-slot selections for individual vehicle drivers, the GSD CarPass Depot App adapts a software-centric QR code permit verification framework specifically for institutional campus management under the University of Iloilo's General Services Department (GSD). Both studies reinforce the core principle that digital, scan-based validation eliminates administrative bottlenecks at security entry points while maintaining secure, real-time access logs. However, the current study tailors its solution to institutional administrative operations by implementing automated email permit delivery, real-time total parking slot occupancy tracking based on gate scans, permit renewal/reactivation workflows, and role-based access control (RBAC) customized across Admins, Staff, and Security Guards.

In an international context, Niketh et al. [1] investigated an intelligent QR code–based university parking management system featuring enhanced security and real-time navigation among university campus motorists, security attendants, and facility managers at SRM Institute of Science and Technology in Kattankulathur, India. The researchers conducted the study to solve severe

campus traffic bottlenecks, unauthorized parking incidents, and search-time delays caused by conventional manual gate-checking and unmonitored parking allocations during peak academic hours. Utilizing an applied, quantitative research and software-engineering design, the study integrated mobile QR authentication with a cloud backend to automate slot reservation and entry verification, revealing that transitioning to a QR-driven digital framework drastically reduced vehicle processing times, eliminated entry queue delays, and improved overall campus security monitoring.

This aligns with the current study's premise that replacing paper-based procedures and manual physical sticker checks with a centralized, QR code-based digital vehicle management system substantially improves administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Niketh et al. [1] prioritized mobile-app dynamic slot reservations, indoor real-time navigation mapping, and direct driver-facing interfaces, the GSD CarPass Depot App adapts a software-centric QR permit framework specifically designed for the General Services Department (GSD) at the University of Iloilo. Both studies reinforce the core principle that digital scan-based verification eliminates administrative bottlenecks at campus gates. However, the current study diverges by incorporating a centralized web application built with Flask Python, automated email distribution of approved QR permits directly to vehicle owners, permit renewal/reactivation features, and role-based access control (RBAC) specifically tailored for Admins, Staff, and Security Guards to addressthe operational inefficiencies, paper-based application backlogs, gate congestion, and lack of real-time parking monitoring facing the university's GSD administration.

In an international context, Hoq et al. [23] investigated a QR-code–based smart car parking system among urban motorists, parking facility operators, and system administrators in Dhaka, Bangladesh. The study was conducted to address the acute urban challenges of rising vehicle

density, severe gate congestion, inefficient manual slot searching, and time-consuming paper ticketing common in traditional parking management. The researchers utilized an experengineering imental and quantitative prototyping method to develop a web-enabled smart parking architecture that generates dynamic QR codes for automated entry/exit authentication, microcontrollers for physical barrier control, and database logging for real-time occupancy management. The study revealed that digitizing access validation via quick response (QR) scanning drastically accelerated vehicle check-in and check-out processing times, eliminated manual paper-based record errors, and provided accurate, real-time monitoring of available parking spaces.

This aligns with the current study's premise that replacing paper-based procedures and manual physical sticker checks with a centralized, QR code-based digital vehicle management system substantially improves administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Hoq et al. [23] prioritized hardware-integrated physical parking slot monitoring and automated check-in/check-out processing via dynamic mobile QR scanning, the current study diverges by incorporating a centralized web-based administrative system built with Flask Python, automated email distribution of vehicle passes, and role-based access control (RBAC) specifically tailored for Admins, Staff, and Security Guards to address the operational inefficiencies, administrative delays, and traffic buildup caused by the manual car pass process of the General Services Department (GSD) at the University of Iloilo

In an international context, Ingole et al. (2024) investigated a QR code–based smart car parking system among urban vehicle motorists, parking facility operators, and gate security personnel in India. The primary motivation for conducting the study stemmed from the operational friction inherent in conventional parking facilities—specifically, severe physical queue delays at entry and exit gates, frequent human data-entry errors, physical paper ticket loss, and the excessive

administrative overhead associated with manual logbook management. Furthermore, traditional manual approaches prevented administrators from gaining real-time visibility into actual space availability, leading to inefficient space utilization and prolonged searching times for drivers. To resolve these challenges, the study utilized an experimental engineering and applied system development research design to architect, integrate, and evaluate a web-enabled smart parking application that generates dynamic Quick Response (QR) codes containing encoded vehicle and driver identification details for automated gate verification. The empirical findings revealed that replacing manual paper ticketing with a scan-based QR code framework reduced vehicle processing times at entry and exit checkpoints by over 60%, eliminated human record-keeping errors, and provided facility managers with reliable, automated tracking of real-time parking slot occupancy. [24]

This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with an automated, QR code-based digital vehicle management system substantially enhances operational efficiency, streamlines access verification, and reduces traffic congestion at entry and exit points. While Ingole et al. (2024) focused primarily on commercial urban parking structures and public driver reservation interfaces, the GSD CarPass Depot App adapts a similar scan-based QR verification architecture tailored specifically to the administrative and operational environment of an academic campus—the General Services Department (GSD) at the University of Iloilo. Both studies reinforce the foundational principle that digitizing vehicle validation at entry points eliminates operational bottlenecks and establishes secure, auditable access logs. However, the current study diverges by incorporating automated email permit distribution directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) customized

across Admins, Staff, and Security Guards, and real-time total campus parking occupancy tracking based on gate scans.

In an international context, Jabbar et al. [2] investigated an Internet of Things (IoT) and Raspberry Pi–based smart parking management system among university drivers, security personnel, and campus facility administrators at a smart university campus setting in Malaysia. The primary motivation for conducting the study stemmed from the persistent operational friction and inefficiencies inherent in conventional campus parking infrastructure—specifically, the severe traffic bottlenecks caused by manual vehicle identification at campus gates, driver frustration and lost time spent searching for unmonitored parking slots, labor-intensive manual gate operations, and the total lack of a centralized real-time monitoring mechanism for campus authorities. To resolve these challenges, the study utilized an experimental hardware-software engineering and quantitative system evaluation research design to architect, assemble, and evaluate a comprehensive smart campus parking platform. Their solution integrated a Raspberry Pi microcontroller, individual physical ultrasonic and infrared slot sensors, camera-based license plate recognition (LPR), automated servo barrier gates, and a centralized web application interface for real-time slot tracking and gate control. The empirical findings revealed that replacing traditional manual check-in procedures with an automated, IoT-driven access control and monitoring system drastically reduced vehicle processing delays at entry checkpoints, virtually eliminated parking search times for drivers, and provided campus authorities with exact, real-time parking space utilization metrics. This aligns with the current study's premise that replacing paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Jabbar et al. [2] prioritized hardware-heavy IoT infrastructure—utilizing physical Raspberry Pi

controllers, individual slot ultrasonic sensors, camera-based license plate recognition, and automated servo barrier gates—the GSD CarPass Depot App adapts a software-centric, highly cost-effective web application framework tailored specifically to the administrative needs of the University of Iloilo's General Services Department (GSD). Both studies reinforce the foundational principle that digitizing vehicle validation at entry points and centralizing access records optimizes campus traffic flow, minimizes physical workload for security personnel, and eliminates operational bottlenecks. However, the current study diverges by incorporating automated QR code permit delivery via email directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on security guard gate scans—providing an accessible, sustainable, and efficient software-driven solution without requiring high-cost physical sensor hardware deployment.

In an international context, Ramdani et al. [9] investigated a web-based parking management system utilizing QR codes and real-time notifications among vehicle drivers, security personnel, and parking facility managers in Indonesia. The primary motivation for conducting the study was to resolve the persistent operational friction and systemic inefficiencies inherent in traditional manual parking management—specifically, prolonged gate processing times caused by physical paper ticket distribution, frequent human data-entry errors in manual logbooks, the complete lack of instant notification mechanisms for drivers, and the absence of a real-time tracking dashboard for facility administrators to monitor space occupancy. To address these operational issues, the study utilized an applied software engineering and system development research design, constructing and evaluating a web application that generates dynamic QR codes for vehicle identity validation, records digital entry and exit logs, and broadcasts instant real-time notifications regarding parking transactions, space status, and duration alerts. The empirical findings of the

study revealed that replacing manual paper ticketing with dynamic QR code authentication significantly accelerated vehicle gate processing times, eliminated physical record-keeping errors, and substantially improved both user and administrator situational awareness through automated operational alerts and centralized data visibility.

This aligns with the current study's premise that replacing paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Ramdani et al. [9] prioritized instant messaging and mobile notification mechanisms to alert drivers directly regarding parking status, entry/exit timestamps, andduration warnings, the GSD CarPass Depot App adapts a software-centric web application specifically tailored to the administrative workflow and organizational structure of the University of Iloilo's General Services Department (GSD). Both studies reinforce the foundational principle that digital scan-based verification eliminates gate bottlenecks while centralizing access records in real time. However, the current study diverges by focusing on institutional vehicle pass management rather than public commercial parking, incorporating automated email permit delivery directly to vehicle owners upon pass generation, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) customized across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on security guard gate scans.

In a foreign context, Remelhe et al. [3] investigated a crowdsourcing- and gamification-based mobile parking management solution among university students, faculty members, administrative staff, and daily campus drivers at the Instituto Politécnico de Viana do Castelo in Viana do Castelo, Portugal. The primary motivation for conducting the study was to mitigate the severe environmental, economic, and operational issues caused by inefficient campus parking

infrastructures—specifically, excessive vehicle search times, recurring gate and aisle traffic congestion, increased fuel consumption, and heightened carbon dioxide (CO2) emissions resulting from drivers idling or circling campus premises while searching for unmonitored open spaces. Recognizing that high hardware deployment and maintenance costs often prevent institutions from installing physical slot sensors across all parking areas, the authors sought to test whether a software-based, community-driven reporting approach could effectively manage campus parking dynamics. To address these challenges, the study utilized an applied software engineering development and descriptive-quantitative user evaluation research design to design, develop, and test a cross-platform React Native mobile application that relies on user-submitted crowdsourced reports and gamification incentives (such as user points, badges, and leaderboards) to maintain and display real-time parking availability by rows and zones rather than individual sensor-monitored spaces. The empirical findings of the study revealed that utilizing a crowdsourced mobile system reduced average campus vehicle search times by approximately 50.75%, corresponding directly to an equivalent 50.75% reduction in campus vehicle carbon emissions, while substantially improving driver convenience, traffic flow, and overall parking space utilization without requiring expensive hardware installations or continuous physical maintenance.

This aligns with the current study's premise that replacing paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Remelhe et al. [3] prioritized a user-driven crowdsourcing model with gamification mechanisms to track row-level slot availability and minimize vehicle searching emissions, the GSD CarPass Depot App adapts a software-centric web application framework tailored specifically to the administrative workflow of the General Services Department (GSD) at the University of Iloilo. Both studies reinforce the

foundational principle that digitizing campus vehicle monitoring optimizes traffic flow and improves operational efficiency without relying on expensive physical hardware deployments. However, the current study diverges by focusing on administrative access control and permit lifecycle management through automated QR code permit delivery via email directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) customized across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy generated automatically from security guard gate scans.

In an international context, Vallikannu et al.[8]investigated an automated vehicle entry management system using image processing and computer vision techniques among security personnel, facility managers, and motorists in India. The primary motivation for conducting the study was to eliminate the operational bottlenecks and security vulnerabilities present in conventional manual access control—specifically, slow gate clearance caused by manual driver logs, physical ticket validation, unauthorized vehicle entries, and the heavy reliance on human guards to visually inspect vehicle license plates at entry checkpoints. To resolve these challenges, the study utilized an experimental engineering and computer vision system development research design, constructing and evaluating an automated framework that captures vehicle images at entry gates, applies Automatic License Plate Recognition (ALPR) via image processing algorithms, and matches license plate strings against an authorized database to grant or deny access automatically. The empirical findings revealed that implementing image processing and automated license plate recognition drastically reduced entry gate verification delays, minimized human error in record-keeping, and provided a highly secure, automated record of all entering and exiting vehicles.

This aligns with the current study's premise that replacing paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access

verification, and reduces traffic congestion at campus entry points. While Vallikannu et al. [8] prioritized computer-vision-driven License Plate Recognition (ALPR) cameras and image processing hardware at entry gates to automate vehicle access, the GSD CarPass Depot App adapts a software-centric web application specifically tailored to the administrative workflow of the General Services Department (GSD) at the University of Iloilo. Both studies reinforce the foundational principle that digitizing vehicle authentication at access checkpoints eliminates physical logbook errors and speeds up gate clearance. However, the current study diverges by utilizing lightweight, cost-effective QR code scanning on mobile devices, automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on security guard gate scans without requiring expensive camera hardware or image processing infrastructure.

In a foreign context, Brown and Hong (2019) investigated the design and implementation of an automated smart campus parking system among faculty members, students, administrative staff, and visitors at Carson-Newman University in Jefferson City, Tennessee. The primary motivation for conducting the study stemmed from the widespread daily frustration, extensive time wastage, traffic congestion, and environmental degradation caused by motorists endlessly circling university grounds searching for open parking spaces under traditional, unmonitored campus parking management practices.

To resolve these recurring administrative and logistical challenges, the study utilized an applied systems design, database architecture development, and prototype evaluation research design to conceptualize and construct a cross-platform mobile application integrated with a real-time Google Firebase cloud database. Their developed solution allowed drivers to view current campus parking availability, reserve specific parking spaces in advance based on their individual academic class

schedules, and verify their physical arrival at designated stalls by scanning unique Quick Response (QR) codes posted at individual parking spots using their smartphones. The developmental findings of the study revealed that implementing a cloud-based, reservation-driven, and QR-authenticated parking system successfully streamlined campus parking organization, drastically reduced searching times for drivers, alleviated daily campus commute stress, and minimized unnecessary vehicle idling emissions across university property.

This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Brown and Hong (2019) focused heavily on individual stall-level reservations tied to student and faculty class schedules along with spot-specific QR code scanning at individual parking bays, the GSD CarPass Depot App adapts a software-centric administrative framework specifically tailored to the overall vehicle permitting and gate control workflows of the General Services Department (GSD) at the University of Iloilo. Both studies reinforce the foundational principle that digitizing vehicle authentication via QR code scanning eliminates physical logbook delays, optimizes campus traffic flow, and centralizes administrative record-keeping. However, the current study diverges by focusing on university gate access management and permit lifecycle administration rather than individual stall-level driver reservations, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on security guard gate scans.

In an international context, Buian et al. (2024) investigated an efficient QR code–based parking management system among urban motorists, parking facility operators, and system administrators.

The primary motivation for conducting the study was to eliminate the operational bottlenecks, excessive driver search times, physical paperwork, human record-keeping errors, and administrative friction inherent in traditional manual parking management systems. To resolve these operational challenges, the study utilized an experimental engineering, IoT integration, and system development research design, constructing and evaluating an automated framework that pairs Quick Response (QR) code authentication with an ESP8266 microcontroller, infrared sensors, and a web-based database to allocate spaces dynamically and verify vehicles upon entry and exit. The empirical findings revealed that scanning unique QR codes for vehicle authentication drastically reduced gate processing delays, minimized manual labor, eliminated logbook errors, and provided real-time visibility into available parking slots This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances operational efficiency, streamlines access verification, and reduces traffic congestion at campus entry points. While Buian et al. (2024) focused on commercial parking spaces with physical hardware integration—specifically utilizing ESP8266 microcontrollers and infrared slot sensors for automated real-time slot assignments—the GSD CarPass Depot App adapts a software-centric web application framework specifically tailored to the administrative workflow of the General Services Department (GSD) at the University of Iloilo. Both studies reinforce the foundational principle that digitizing vehicle authentication via QR code scanning eliminates physical logbook delays, optimizes traffic flow, and centralizes administrative record-keeping. However, the current study diverges by targeting institutional vehicle pass lifecycle management without requiring expensive hardware sensor deployments, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and

software-calculated total campus parking occupancy based on security guard gate scans. In an international context, Demapure et al. (2024) investigated an intelligent vehicle access management system titled SecurePass among facility security personnel, motorists, and system administrators. The primary motivation for conducting the study was to eliminate the operational bottlenecks, security vulnerabilities, and administrative friction inherent in traditional manual access control—specifically, slow gate entry caused by handwritten driver logbooks, unverified visitor access, lack of real-time vehicle movement tracking, and heavy reliance on human guards to inspect credentials visually.

To resolve these operational challenges, the study utilized an experimental engineering and software system development research design, constructing and evaluating an automated framework that integrates Quick Response (QR) code technology, cloud-based database management, and mobile scanning applications to authenticate authorized vehicles, log timestamped entry/exit data in real time, and trigger automated gate notifications. The empirical findings revealed that digitizing vehicle access control through intelligent QR-based verification drastically reduced gate processing delays, eliminated human logging errors, provided immediate administrative auditing, and significantly strengthened overall security control. [6]

This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While SecurePass prioritized an automated access control architecture tailored primarily for general intelligent access logging and physical gate integration, the GSD CarPass Depot App adapts a software-centric administrative framework specifically tailored to the overall vehicle permitting, gate validation, and institutional workflows of the General Services Department (GSD) at the University of Iloilo.

Both studies reinforce the foundational principle that replacing manual paper logbooks with QR code digital tracking centralizes administrative oversight, optimizes gate clearance speeds, and elevates campus security. However, the current study diverges by focusing on university-wide vehicle pass lifecycle management rather than just access logging, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier. Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on real-time mobile security guard gate scans.

In an international context, Raman et al.[4]investigated intelligent parking management systems for real-time space availability estimation among urban motorists, commercial parking managers, and facility administrators. The primary motivation for conducting the study was to mitigate severe urban traffic congestion, reduce driver search time and fuel wastage, and eliminate the operational inefficiencies of traditional manual parking management, where drivers circle facilities aimlessly due to a lack of live parking visibility. To resolve these operational and environmental challenges, the study utilized an experimental engineering, Internet of Things (IoT) integration, and machine learning framework research design. The technology used in their solution combined IoT camera sensors and ultrasonic physical detectors with computer vision algorithms, Convolutional Neural Networks (CNN), and cloud-based data processing to continuously monitor parking bay status, predict space availability, and transmit live occupancy analytics to drivers in real time. The empirical findings revealed that implementing machine learning-driven visual detection and IoT sensor networks accurately estimated real-time parking space availability, drastically minimized driver search times, optimized parking lot capacity, and significantly reduced carbon emissions caused by idling vehicles. This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative

efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Raman et al. [4] focused on hardware-heavy sensor networks, computer vision cameras, and predictive machine learning models to detect individual stall occupancy, the GSD CarPass Depot App adapts a software-centric administrative framework specifically tailored to the overall vehicle permitting, gate validation, and institutional access control workflows of the General Services Department (GSD) at the University of Iloilo.

Both studies reinforce the foundational principle that replacing manual record-keeping with real-time digital monitoring centralizes administrative oversight, optimizes vehicle throughput, and eliminates unnecessary congestion. However, the current study diverges by leveraging lightweight, cost-effective mobile Quick Response (QR) code scanning rather than expensive hardware sensors and artificial intelligence cameras, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on real-time mobile security guard gate scans.

In an international context, Bankar et al. [25] investigated a smart parking management system based on Quick Response (QR) code technology among urban motorists, commercial parking facility operators, and system administrators in India. The primary motivation for conducting the study was to eliminate the chronic traffic congestion, driver time wastage, environmental pollution, and administrative inefficiencies caused by conventional manual parking practices, where drivers waste considerable time searching for open spaces and waiting in line for physical paper tickets.

To resolve these operational challenges, the study utilized an experimental system design, software development, and prototype evaluation research design. The technology used in their solution integrated Android mobile application development, QR code generation and scanning modules, and a centralized web database server to enable drivers to view slot availability, pre-book parking

spaces, generate unique QR code tokens on their smartphones, and scan them at entry/exit points for automated access and digital fee calculation. The empirical findings revealed that implementing a QR-based mobile reservation system drastically reduced entry clearance delays, eliminated paper ticket handling, minimized vehicle searching time, and significantly improved the overall operational efficiency and slot utilization of parking facilities.

This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Bankar et al. [25] focused on commercial parking lot reservations, spot-level pre-booking, and automated fee payment integration via individual smartphone QR generation, the GSD CarPass Depot App adapts a software-centric administrative framework specifically tailored to the overall vehicle permitting, gate validation, and institutional access control workflows of the General Services Department (GSD) at the University of Iloilo.

Both studies reinforce the foundational principle that replacing manual physical logbooks with mobile QR code scanning centralizes administrative oversight, accelerates entry clearance speeds, and eliminates traffic bottlenecks. However, the current study diverges by targeting institutional vehicle pass lifecycle management rather than commercial slot pre-booking, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on real-time mobile security guard gate scans.

In an international context, Tiwari et al. [26] investigated a smart parking system application utilizing Quick Response (QR) code technology among urban motorists, parking attendants, and

facility administrators in India. The primary motivation for conducting the study was to alleviate the daily frustration, severe traffic bottlenecks, driver time wastage, and environmental pollution caused by traditional manual parking systems, where drivers waste time searching for open slots and attendants rely on physical ticketing and manual paper ledgers.

To resolve these operational and environmental challenges, the study utilized an applied software engineering development and system evaluation research design. The technology used in their solution centered on an Android mobile application integrated with a centralized database and real-time QR code scanning capabilities, allowing drivers to view available parking slots, book spaces in advance, and scan QR codes at parking entry and exit points for automated fee calculation and access validation. The empirical findings revealed that implementing a mobile QR-based parking application significantly reduced parking search times, eliminated manual paper-based errors and ticketing delays, optimized overall parking space utilization, and provided a seamless, contact-free experience for drivers.

This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Tiwari et al. [26] focused on an Android application tailored for public commercial parking spaces, advance slot reservations, and automated billing, the GSD CarPass Depot App adapts a software-centric web application framework specifically tailored to the institutional vehicle permitting, gate validation, and access control workflows of the General Services Department (GSD) at the University of Iloilo.

Both studies reinforce the foundational principle that replacing manual physical record-keeping with QR-based digital verification centralizes administrative oversight, speeds up entry clearance,

and reduces vehicular congestion. However, the current study diverges by focusing on university-wide car pass lifecycle management rather than commercial slot booking and fee collection, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on real-time mobile security guard gate scans.

In an international context, Haider Rizvi et al.[27]investigated a cloud-based smart parking system among urban drivers, commercial parking facility operators, and system administrators. The primary motivation for conducting the study was to alleviate the widespread daily frustration, excessive time loss, fuel waste, and traffic congestion caused by traditional, unmonitored parking management, where motorists manually search for open parking spots without prior knowledge of space availability.

To resolve these recurring urban mobility challenges, the study utilized an experimental system design and Cloud-IoT prototype development research design. The technology used in their solution integrated NodeMCU (ESP8266) microcontrollers, ultrasonic proximity sensors placed at individual parking slots, a Firebase real-time cloud database, and a cross-platform mobile application, allowing drivers to view live slot occupancy, reserve available spaces remotely, and receive automated parking direction updates. The empirical findings revealed that implementing a cloud-connected smart parking infrastructure significantly reduced driver searching and queuing times, maximized parking lot space utilization, eliminated manual slot monitoring errors, and streamlined overall facility management through real-time cloud synchronization.

This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up

access verification, and reduces traffic congestion at campus entry points. While Haider Rizvi et al.[27]focused on hardware-centric Internet of Things (IoT) slot sensors, NodeMCU microcontrollers, and individual stall-level driver reservations, the GSD CarPass Depot App adapts a software-centric administrative framework specifically tailored to the overall vehicle permitting, gate validation, and institutional access control workflows of the General Services Department (GSD) at the University of Iloilo.

Both studies reinforce the foundational principle that replacing manual physical logbooks with cloud-connected digital tracking centralizes administrative oversight, optimizes gate clearance speeds, and eliminates unnecessary campus congestion. However, the current study diverges by leveraging lightweight, cost-effective mobile Quick Response (QR) code scanning rather than installing individual physical slot sensors and microcontrollers, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on real-time mobile security guard gate scans.

In an international context, Tripathi[28]investigated a smart vehicle parking system using IoT among urban drivers, parking lot managers, and facility administrators. The primary motivation for conducting the study was to solve the severe urban traffic bottlenecks, excessive driver searching delays, unnecessary fuel consumption, and operational friction caused by traditional, unmonitored parking management, where motorists manually look for available spaces without real-time information.

To resolve these pressing urban mobility issues, the study utilized an experimental engineering and IoT prototype system development research design. The technology used in the solution integrated physical infrared (IR) sensors installed at individual parking stalls, an Arduino microcontroller, Wi-Fi modules (ESP8266), and a cloud-based web application, enabling the

system to detect vehicle presence at individual slots continuously, process occupancy data in real time, and display slot availability to drivers via a web interface. The empirical findings revealed that implementing an automated IoT-driven parking system significantly decreased vehicle queueing and searching times, maximized space allocation efficiency, eliminated manual oversight errors, and reduced vehicle emissions by preventing unnecessary circling within parking facilities. This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points.

While Tripathi [28] prioritized hardware-driven IoT components, such as physical IR sensors and microcontroller boards at individual parking bays for real-time stall detection, the GSD CarPass Depot App adapts a software-centric administrative framework specifically tailored to the overall vehicle permitting, gate validation, and institutional access control workflows of the General Services Department (GSD) at the University of Iloilo.

Both studies reinforce the foundational principle that replacing manual physical logbooks with digital tracking centralizes administrative oversight, optimizes gate clearance speeds, and eliminates unnecessary campus congestion. However, the current study diverges by leveraging lightweight, cost-effective mobile Quick Response (QR) code scanning rather than installing individual physical slot sensors and microcontroller hardware, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier. Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on real-time mobile security guard gate scans.

In an international context, Prakash (2022) investigated a smart car parking system using IoT among urban drivers, parking lot operators, and facility administrators. The primary motivation

for conducting the study was to alleviate severe urban traffic congestion, minimize driver searching delays and unnecessary fuel consumption, and eliminate the operational inefficiencies of traditional, unmonitored parking management, where motorists manually look for available spaces without real-time information. [29]

To resolve these pressing urban mobility challenges, the study utilized an experimental engineering and IoT prototype system development research design. The technology used in the solution integrated physical infrared (IR) sensors positioned at individual parking bays, microcontrollers, Wi-Fi connectivity modules, and a centralized web or mobile application interface. This setup continuously detects vehicle presence at specific parking slots, transmits live occupancy status updates to cloud servers, and displays available parking spaces to motorists in real time. The empirical findings revealed that implementing an automated IoT-driven parking framework significantly decreased vehicle searching and queueing times, maximized overall facility space utilization, eliminated manual record-keeping errors, and reduced vehicle carbon emissions by preventing unnecessary idling.

This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Prakash (2022) prioritized hardware-driven IoT components—specifically installing physical IR sensors and microcontroller boards at individual parking stalls for real-time bay detection—the GSD CarPass Depot App adapts a software-centric administrative framework specifically tailored to the overall vehicle permitting, gate validation, and institutional access control workflows of the General Services Department (GSD) at the University of Iloilo.

Both studies reinforce the foundational principle that replacing manual physical logbooks with digital tracking centralizes administrative oversight, optimizes gate clearance speeds, and eliminates unnecessary campus congestion. However, the current study diverges by leveraging lightweight, cost-effective mobile Quick Response (QR) code scanning rather than installing individual physical slot sensors and microcontroller hardware, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on real-time mobile security guard gate scans.

2.2 Local Studies

In a local context, Luciano et al. (2023) investigated a web-based parking lot management system titled PARADAJuan among drivers, parking attendants, and facility administrators at Bulacan State University in the Philippines. The primary motivation for conducting the study stemmed from the severe operational friction and administrative inefficiencies inherent in conventional manual parking operations—specifically, prolonged gate processing delays, manual paper ticket handling, lack of real-time slot availability visibility, human record-keeping errors in physical logbooks, and driver frustration caused by searching aimlessly for open spaces. [30]

To resolve these challenges, the study utilized an agile software engineering development and descriptive-quantitative system evaluation research design, designing and constructing a web application using multi-paradigm programming languages to automate driver registration, parking slot tracking, and transaction logging, followed by a user evaluation structured around the ISO/IEC 25010 software quality standard. The empirical findings revealed that automating parking management through a centralized web platform drastically reduced vehicle processing times, eliminated manual paper-based record errors, optimized overall parking space utilization, and achieved high performance ratings across functionality, usability, and reliability metrics

processing times, eliminated manual paper-based record errors, optimized overall parking space utilization, and achieved high performance ratings across functionality, usability, and reliability metrics.

This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Luciano et al. (2023) prioritized a general web-based parking lot platform centered on real-time visual slot mapping, stall-by-stall space allocation, and multi-paradigm software engineering for localized parking bays, the GSD CarPass Depot App adapts a software-centric administrative framework specifically tailored to the overall vehicle permitting, gate validation, and institutional access control workflows of the General Services Department (GSD) at the University of Iloilo.

Both studies reinforce the foundational principle that replacing manual physical record-keeping with digital web architecture centralizes administrative oversight, optimizes gate clearance speeds, and elevates campus security. However, the current study diverges by focusing on university-wide vehicle pass lifecycle administration and gate-level validation rather than localized parking slot reservations, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on security guard gate scans.

In a local institutional context, Palasan et al. (2025) investigated the design and implementation of a web-based campus vehicle gate pass management system with Optical Character Recognition (OCR) document verification among campus security personnel (OCSSS), administrative staff, and campus vehicle owners at Caraga State University in Butuan City, Philippines. The study

utilized an applied software engineering research design paired with a System Usability Scale (SUS) quantitative evaluation method, revealing that digitizing gate pass requests and automating credential validation (verifying OR/CR documents and driver's licenses via OCR) significantly eliminated manual application backlogs, reduced administrative verification errors, and enhanced campus security monitoring through centralized data tracking. [31]

This aligns with the current study's premise that replacing traditional paper-based procedures and manual physical sticker checking with a centralized digital vehicle management system substantially improves administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Palasan et al. (2025) prioritized OCR-driven document parsing to automate the verification of physical vehicle registration documents (OR/CR) and driver's licenses during the application stage, the GSD CarPass Depot App focuses on streamlining campus entry gate throughput and permit delivery via QR code technology, featuring automated email distribution of generated QR permits, real-time parking slot occupancy tracking based on entry/exit gate scans, and role-based access control (RBAC) across Admins, Staff, and Security Guards to address the operational inefficiencies, manual paper-based car pass procedures, gate congestion, and lack of real-time parking monitoring facing the General Services Department (GSD) at the University of Iloilo.

In a local context, Luzuriaga et al.[32]investigated a parking occupant management system utilizing Quick Response (QR) code solutions secured with the Advanced Encryption Standard (AES) algorithm among campus drivers, parking managers, and security staff in the Philippines. The primary motivation for conducting the study stemmed from the operational bottlenecks and security risks prevalent in traditional manual vehicle logs—specifically, slow gate validation, vulnerability to counterfeit or unauthorized physical access permits, paper wastage, and human error during gate record-keeping.

To address these challenges, the study utilized an applied software development and descriptive-quantitative system evaluation research design, engineering a web-based platform that generates encrypted, time-stamped QR codes using 128-bit or 256-bit AES cryptographic algorithms to authenticate vehicle entry and exit while continuously tracking slot occupancy. The empirical evaluation revealed that implementing AES-encrypted QR code scanning drastically accelerated gate processing speeds, eliminated permit falsification, prevented unauthorized vehicle access, and provided accurate real-time data on parking occupancy without administrative delay.

This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Luzuriaga et al.[32]prioritized data-level cryptographic security via AES encryption for single-scan QR validation and localized parking slot allocation, the GSD CarPass Depot App adapts a software-centric administrative framework specifically tailored to the overall vehicle permitting, gate validation, and institutional access control workflows of the General Services Department (GSD) at the University of Iloilo.

Both studies reinforce the foundational principle that digitizing vehicle authentication using QR technology centralizes administrative oversight, optimizes gate clearance speeds, and elevates campus security. However, the current study diverges by focusing on university-wide vehicle pass lifecycle administration rather than specific cryptographic payload encryption, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on real-time security guard gate scans. In a local context, Morata et al. [33] investigated the SEAIT Smart Parking System

designed to enhance efficiency and security in campus vehicle management among students, faculty, staff, and security personnel at the South East Asian Institute of Technology (SEAIT) in the Philippines. The primary motivation for conducting the study was to overcome the severe administrative friction and operational vulnerabilities of traditional manual campus parking systems—specifically, chronic gate entry bottlenecks, human error in physical paper logbooks, lack of real-time parking space availability monitoring, and unmonitored vehicle movements across university premises.

To resolve these operational challenges, the study utilized an applied software development and descriptive-quantitative system evaluation research design, constructing an integrated smart campus parking framework that utilizes RFID technology, ultrasonic sensors, and web-based dashboard monitoring to track slot occupancy in real time and authenticate registered campus vehicles upon arrival. The empirical findings revealed that digitizing vehicle access and slot tracking through an automated smart system drastically reduced gate queueing times, eliminated manual paper-based record-keeping errors, provided accurate real-time visibility of open parking spaces, and significantly elevated overall campus security and administrative oversight.

This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Morata et al. [33] prioritized hardware-driven RFID tags, gate barriers, and ultrasonic physical slot sensors for automated parking bay monitoring, the GSD CarPass Depot App adapts a software-centric administrative framework specifically tailored to the overall vehicle permitting, gate validation, and institutional access control workflows of the General Services Department (GSD) at the University of Iloilo.

Both studies reinforce the foundational principle that replacing manual physical logbooks with digital tracking centralizes administrative oversight, optimizes gate clearance speeds, and elevates campus security. However, the current study diverges by leveraging lightweight, cost-effective mobile QR code scanning rather than expensive hardware sensors and physical RFID readers, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on real-time security guard gate scans.

In an international context, Satpalkar et al.[34]investigated a smart city parking system using a Quick Response (QR) code–based approach among urban drivers, parking attendants, and municipalityfacility administrators. The primary motivation for conducting the study was to address the growing urban traffic congestion, prolonged search times for open parking spots, fuel inefficiency, and administrative vulnerabilities associated with traditional paper-ticket parking systems—specifically, gate bottlenecks, physical ticket loss, human error in logbook entry, and lack of real-time space tracking.

To resolve these pressing urban infrastructure challenges, the study utilized an experimental software prototyping and system development research design. The technology used in their solution integrated Android mobile applications, dynamic QR code generation algorithms, centralized web servers, and relational database management systems (RDBMS) to generate unique digital QR passes for vehicle registration, authenticate drivers at entry/exit points, track available slots, and calculate parking durations automatically. The empirical findings revealed that implementing a QR code–driven parking architecture drastically accelerated gate authentication speeds, eliminated paper ticket handling, minimized human record-keeping errors, and optimized parking space allocation across busy metropolitan facilities.

This aligns with the current study's premise that replacing traditional paper-based procedures, manual physical sticker verification, and compulsory physical office visits with a centralized digital vehicle management system substantially enhances administrative efficiency, speeds up access verification, and reduces traffic congestion at campus entry points. While Satpalkar et al.[34]prioritized public municipal parking zones, dynamic commercial slot booking, and time-based fee billing algorithms via driver-side Android devices, the GSD CarPass Depot App adapts a software-centric administrative framework specifically tailored to the overall vehicle permitting, gate validation, and institutional access control workflows of the General Services Department (GSD) at the University of Iloilo.

Both studies reinforce the foundational principle that replacing manual physical logbooks with Quick Response (QR) code digital authentication centralizes administrative oversight, optimizes gate clearance speeds, and eliminates unnecessary congestion. However, the current study diverges by focusing on university-wide vehicle pass lifecycle management rather than public commercial parking booking or billing, incorporating automated email permit delivery directly to vehicle owners, permit renewal and reactivation workflows, three-tier Role-Based Access Control (RBAC) across Admins, Staff, and Security Guards, and software-calculated total campus parking occupancy based on real-time mobile security guard gate scans.

2.3 Related Literature

Managing campus vehicular traffic, access control, and parking permits presents a critical, multi-faceted operational challenge for modern educational institutions. As vehicle volume among faculty, staff, students, and visitors continues to grow, traditional administrative methods—such as paper-based application forms, manual sticker verification, physical gate checks, and handwritten logbooks—inevitably create severe operational bottlenecks. These manual workflows result in extended gate clearance delays, peak-hour campus traffic gridlock, a complete lack of

real-time occupancy monitoring, and heightened security vulnerabilities regarding counterfeit permits or unauthorized entry. To address these persistent challenges, academic institutions are increasingly adopting digital management applications and web-based access architectures to streamline gate operations, automate credential verification, and elevate centralized administrative oversight. This comprehensive related literetu twenty-five key local and international studies to establish the theoretical, technological, and empirical foundations of digital vehicle management systems. The literature is systematically categorized into four major thematic areas derived from the core operational problems encountered in institutional vehicle administration: Traffic Congestion & Gate Bottlenecks, Manual Operational Inefficiencies & Delay, Real-Time Monitoring & Centralized Data Management, and Security, Authentication, & Access Control. To rigorously examine the technological evolution and scientific methodologies behind these solutions, the related studies within each theme are grouped by their specific research designs—ranging from experimental prototyping and Cloud-IoT frameworks to software engineering evaluations—and arranged chronologically from oldest to latest.

Together, these studies demonstrate a clear academic consensus on the necessity of replacing physical, paper-based gate checks with automated, digital validation frameworks. While many existing studies rely on costly hardware deployments—such as IoT slot sensors, automated barrier gates, or automatic license plate recognition (ALPR) cameras—the body of literature provides direct justification for the development of the GSD CarPass Depot App at the University of Iloilo. By adopting a lightweight, software-centric approach utilizing Python Flask, MySQL central databases, automated email permit delivery, and mobile QR code validation, the system applies the empirical findings established in scholarly research to overcome manual operational delays through an affordable, scalable, and highly secure institutional solution.

Satpalkar et al. [34] used an experimental software prototyping and system development research design to demonstrate how transitioning from manual sticker verification to dynamic quick response (QR) code passes effectively eliminates queuing bottlenecks at high-density access gates. Their study highlights that manual gate checks create artificial delays due to slow visual inspections, whereas dynamic software verification automates driver identity validation in milliseconds. The researchers demonstrated that digital pass verification removes human verification error while significantly smoothing vehicle inflow during peak arrival hours. By replacing physical permits with dynamically generated mobile credentials, the research team successfully established a scalable software-driven baseline for automated perimeter security. This prototype proved that automated digital validation drastically reduces entry gate friction without requiring expensive mechanical gates. Building on this framework, Brown and Hong (2019) implemented an applied systems design, database architecture development, and prototype evaluation research design at Carson-Newman University. By linking a cloud-connected mobile app to live academic schedules, they proved that synchronized QR scanning not only speeds up gate entry but also reduces campus-wide traffic gridlock and vehicular idling emissions. Their research validated that software-driven scheduling integration prevents bottlenecks by systematically distributing vehicle entry clearance across designated time slots. The study's prototype evaluation confirmed that real-time cloud database synchronization prevents entry lane congestion by validating driver access permissions instantly upon arrival. Furthermore, the system drastically minimized vehicle idling, demonstrating that digital scheduling directly cuts carbon emissions near institutional entrances.

Hoq et al.[23]similarly utilized an experimental and quantitative prototyping method to evaluate driver check-in throughput. Their findings validated that replacing physical tickets with instant quick-response scans at entry and exit points drastically accelerates vehicle processing speeds,

preventing traffic back-ups from spilling onto surrounding public roadways during peak operating hours. The empirical data confirmed that digital access validation significantly cuts check-in latency, allowing institutions to maximize gate capacity without physically expanding gate lanes. Through rigorous field trials, the authors observed that scan-based validation maintains consistent clearance rates even during heavy arrival surges. Consequently, their prototyping method proved that software-based entry workflows offer a highly cost-effective remedy for urban and institutional traffic congestion.

Felisilda and Narisma[35]conducted an early foundational study using a software engineering design structured around the Waterfall Model, integrating physical microcomputers and hardware sensors to automate gate clearance. Their research established that replacing manual paper logbooks with automated barrier triggers significantly increases vehicle entry rates, proving that gate delays are primarily caused by slow human logging procedures. The study laid the early groundwork for automated campus access by illustrating how physical hardware controls perform best when linked directly to electronic verification databases. Despite hardware limitations of that era, the authors successfully demonstrated that embedded systems can reliably automate gate credentials. Their foundational methodology established that removing pen-and-paper record-keeping is the single most critical step toward solving entrance bottlenecks.

Decades later, Jabbar et al.[2]advanced this concept through an experimental hardware-software engineering and quantitative system evaluation research design. Deploying Raspberry Pi microcontrollers alongside automated barrier gates on an academic campus, they demonstrated that combining hardware triggers with digital database checks prevents vehicular congestion along main campus arteries, ensuring smooth traffic flow even during high-volume arrival times. Their system quantitative evaluation confirmed that hardware automation cuts gate queuing delays to a minimum. The researchers conducted extensive stress-testing to measure hardware response times

under extreme peak-load traffic scenarios. Their findings proved that localized edge-computing microcontrollers seamlessly sync with backend databases to deliver split-second gate opening approvals. This line of engineering research proves that physical gateinfrastructure operates far more effectively when manual logbooks are completely removed from the loop. Integrating physical controls directly with central verification systems drastically reduces stopping time at access points. However, while physical barriers provide effective perimeter control, the underlying research highlights that the primary driver of throughput speed is the rapid digital validation process running in the background. When hardware controllers are synchronized with optimized software backends, gate clearance operational capacity expands dramatically. Ultimately, these combined engineering methodologies emphasize that hardware reliability must always be paired with streamlined digital validation algorithms to eliminate physical queue formation.

Remelhe et al.[3]executed an applied software engineering development and descriptive-quantitative user evaluation research design at Instituto Politécnico de Viana do Castelo. By combining real-time parking tracking with mobile user feedback, they revealed that digital access management cuts campus vehicle search times and reduces idling-related carbon emissions by 50.75%. Their evaluation showed that driver uncertainty directly contributes to gate and lane congestion, which dynamic software monitoring actively resolves. The research team gathered detailed quantitative metrics tracking vehicle dwell times before and after implementing the mobile software solution. Their statistical analysis confirmed a massive reduction in campus driving delays, validating the ecological and operational benefits of digital parking guidance.

Following a similar quantitative path, Niketh et al.[1]applied a software-engineering and quantitative evaluation design at SRM Institute of Science and Technology. Their study confirmed that deploying guard-operated mobile QR authentication backed by cloud servers completely removes gate arrival queues during morning rush hours, proving that lightweight software apps

can achieve the same rapid gate throughput as expensive physical barrier hardware. Quantitative field trials verified that mobile scanning by gate personnel drastically cuts entry clearance times. By recording exact vehicle processing timestamps at entry checkpoints, the researchers established that guard-side mobile app validation processes vehicles in a fraction of the time required by traditional methods. This empirical validation verified that software applications eliminate the need for heavy capital expenditures on mechanical barrier gates.

Together, these quantitative studies emphasize that operational congestion is fundamentally an information and verification problem rather than a lack of physical gate space. Equipping gate personnel with rapid mobile scanning tools eliminates queues without requiring heavy infrastructure modifications. Applied software evaluation frameworks consistently prove that digitalizing entry validation yields immediate, measurable reductions in gate traffic congestion. By replacing subjective visual inspections with instant database checks, institutional entrance throughput reaches optimal mathematical efficiency. Furthermore, these quantitative evaluation methodologies offer repeatable benchmarks that allow future researchers to evaluate gate management software performance against real-world traffic flows.

Bankar et al.[25]employed an experimental system design, software development, and prototype evaluation research design to resolve administrative backlogs. Their system paired Android mobile apps with a central web server to generate dynamic digital tokens, proving that eliminating paper ticket issuance completely removes administrative handling delays and manual registration errors. Prototype testing demonstrated that centralizing token data streamlines administrative record management across the entire facility. The authors conducted systematic performance trials comparing paper ticket issuance times against instantaneous digital token generation. Their experimental results proved that digital database entry prevents administrative data loss, manual typo errors, and physical filing backlogs.

Dokania et al.[7]adopted an experimental engineering and system development research design using Flutter and Firebase ML Vision. Their empirical testing showed that replacing physical paper permits with digital QR e-tickets drastically reduces processing times for drivers while eliminating the heavy administrative overhead burdening facility managers. The study proved that automated digital validation removes the labor-intensive filing processes typical of legacy parking setups. By building a cross-platform mobile architecture, the engineering team verified that digital credential creation requires minimal server processing power while offering instant verification capabilities. Their field testing confirmed that administrative personnel save hundreds of operational hours annually by managing credentials digitally.

Buian et al. (2024) applied an experimental engineering, IoT integration, and system development research design, finding that dynamic QR authentication minimizes human intervention and removes paper-based friction from routine driver check-ins. Similarly, Ingole et al. (2024) utilized an experimental engineering and applied system development research design, conducting timing trials that proved scan-based QR authentication cuts entry and exit processing times by over 60% compared to traditional handwritten security logging. Ingole et al. meticulously documented time expenditures across various driver validation steps, showing that manual paper logging creates extreme administrative drag. Their software solution completely digitized the visitor registration pipeline, removing human logging mistakes and providing instant digital receipts for institutional records.

Tiwari et al.[26]utilized an applied software engineering development and system evaluation research design to address operational delays in high-traffic facilities. Their field evaluations proved that replacing physical record books with digital QR validation eliminates human record-keeping errors and drastically speeds up overall vehicle processing times. The empirical evidence confirmed that paper ledger systems are inherently error-prone and introduce severe administrative

delays. Through rigorous systematic field testing across multiple facility entry points, the software evaluation team tracked daily administrative error rates. Their findings demonstrated that transitioning to digital database logging reduces administrative record discrepancies to absolute zero.

Sravanthi and Rajender[22]reinforced these findings using an applied software development and experimental quantitative evaluation method. By subjecting their QR parking app to simulated high-traffic loads, they collected empirical data confirming that digital scan validation accelerates gate clearance, eliminates operational bottlenecks, and modernizes institutional vehicle management. Their system evaluation demonstrated consistent time savings across all administrative user groups. The researchers ran stress tests simulating thousands of simultaneous validation requests to evaluate database responsiveness. The quantitative results proved that cloud-hosted database architectures maintain sub-second response times regardless of query volume, proving their resilience for large institutional deployments.

Applied software engineering methodologies consistently demonstrate that replacing paper processes with web or mobile applications eliminates administrative backlogs. Digital record keeping enables instant searchability and automated logging, which physical ledgers cannot provide. These studies validate that software engineering approaches are essential for modernizing institutional vehicle administration. By eliminating physical application forms and manual sticker distribution, organizations achieve near-instantaneous permit approval cycles. Furthermore, software engineering evaluation designs prove that centralized web-based workflows significantly improve data accuracy while cutting operational administrative costs. Luciano et al. (2023) evaluated PARADAJuan, a web-based parking management system at Bulacan State University, using an agile software engineering development and descriptive-quantitative evaluation framework structured around ISO/IEC 25010 software quality standards. Their findings

demonstrated that automating vehicle registration and transaction logging significantly improves functional suitability and reduces operational delays across campus. Adhering to international software standards ensured that the platform met strict usability and efficiency metrics. The agile methodology enabled continuous iterative testing with campus stakeholders, directly addressing operational paint points during early sprint cycles. ISO/IEC 25010 evaluations confirmed superior scores across system reliability, security, and performance efficiency. Palasan et al. (2025) addressed vehicle pass application backlogs at Caraga State University by pairing an applied software engineering research design with a System Usability Scale (SUS) quantitative evaluation. Their research validated that combining web-based pass requests with Optical Character Recognition (OCR) eliminates administrative application backlogs, reduces verification errors, and delivers high end-user satisfaction. The usability scores confirmed that intuitive digital interfaces encourage rapid user adoption and minimize administrative queries. By collecting quantitative SUS feedback from both vehicle owners and administrative officers, the researchers proved that clear user-interface design directly accelerates form processing. The system's automated document extraction further reduced human data entry workload.

Standardized usability frameworks prove that system adoption depends directly on software clarity and interface efficiency. Evaluating platforms against recognized frameworks like ISO/IEC 25010 and SUS guarantees that administrative bottlenecks are truly resolved rather than shifted to complex software interfaces. These methodologies confirm that user-centered software design is vital for eliminating institutional delays. When administrative interfaces are designed intuitively, processing efficiency improves across all user demographics. Consequently, standardized usability methodologies provide crucial qualitative and quantitative safeguards, ensuring that newly deployed digital access management systems achieve immediate institutional adoption.

Tripathi[28]investigated real-time facility visibility using an experimental engineering and IoT prototype system development research design. By integrating infrared sensors, Wi-Fi microcontrollers, and a cloud web application, the study proved that centralizing vehicle logs gives administrators instant visibility into live parking availability. The experimental prototype proved that real-time data streaming eliminates manual occupancy checks and visual inspections. Tripathi built and calibrated custom sensor node hardware to monitor stall occupancy automatically. The experimental testing verified that real-time data feeds instantly update central web dashboards, eliminating the need for security personnel to manually count parked vehicles.

Haider Rizvi et al.[27]expanded this architecture using an experimental system design and Cloud-IoT prototype development design. Utilizing NodeMCU microcontrollers and a real-time Google Firebase database, they verified that continuous cloud database synchronization completely eliminates manual slot tracking errors and ensures data accuracy. The research proved that live cloud sync prevents data discrepancies between security gates and central administrative offices. By deploying cloud socket connections, the researchers achieved instantaneous bi-directional communication between field sensors and administrative monitors. Their system evaluations demonstrated that real-time database synchronization remains fully operational under continuous continuous heavy data loads.

Prakash (2022) likewise applied an experimental engineering and IoT prototype system development research design, proving that live dashboard updates stream continuous occupancy metrics that drastically improve space utilization and facility oversight. The empirical findings confirmed that centralized monitoring enables proactive space management. Continuous data collection gives management actionable historical insights for future facility planning. Prakash tested sensor transmission reliability across wide physical distances, proving that wireless networks can sustain stable telemetry feeds across large institutional parking grounds. The study

concluded that real-time visual analytics empower facility managers to optimize space allocations instantly.

Raman et al.[4]evaluated facility management using an experimental engineering, IoT integration, and machine learning framework research design. By pairing camera sensors with Convolutional Neural Networks (CNN), they proved that automated image processing optimizes space capacity management and removes the need for manual guard surveillance. The machine learning framework demonstrated automated vehicle classification and tracking without human intervention. The research team trained deep learning models on extensive video datasets to achieve high license plate and vehicle identification accuracy. Their experimental results proved that AI-driven vision systems can monitor vehicle movements in real time while dynamically updating central administrative databases.

Morata et al.[33]applied an applied software development and descriptive-quantitative system evaluation research design to the SEAIT Smart Parking System. Their research confirmed that combining RFID tags with unified web dashboard monitoring significantly enhances administrative oversight and operational transparency. User evaluation metrics indicated high administrative approval for centralized visual dashboards over fragmented physical reporting. Morata et al. gathered extensive quantitative feedback from facility managers who evaluated the system's live mapping interfaces. The survey metrics proved that centralized data visualization reduces administrative reaction time to space conflicts, significantly elevating institutional operational control.

Ramdani et al.[9]followed with an applied software engineering and system development research design in Indonesia. Their system demonstrated that generating digital entry/exit logs via dynamic QR scans and rendering live analytics on a central dashboard vastly improves administrative situational awareness without requiring complex physical sensor networks. The study proved that

software-driven transaction logging provides complete operational visibility at a fraction of hardwarecosts. By tracking every scan event in a structured relational database, the software dynamically computes live facility occupancy percentages. This software-centric framework proved that intelligent backend logging delivers robust administrative monitoring without hardware maintenance burdens.

Vallikannu et al.[8]addressed gate security using an experimental engineering and computer vision system development research design. Deploying Automatic License Plate Recognition (ALPR) algorithms, they proved that automated database matching eliminates human verification errors and prevents unauthorized vehicles from entering restricted zones. Experimental testing confirmed that optical plate matching instantly flags unregistered vehicles at access points. The researchers conducted rigorous field tests under diverse lighting and weather conditions to measure computer vision accuracy. Their findings demonstrated that automated camera validation stops unauthorized vehicle entry far more reliably than visual checks by security guards.

Demapure et al. (2024) applied an experimental engineering and software system development research design to SecurePass. Their security testing proved that replacing handwritten guard logbooks with cloud-connected QR scanning enhances security auditing, prevents logbook tampering, and creates an unalterable digital paper trail. Security evaluations showed that cloud-backed scanning prevents retrospective log manipulation by unauthorized personnel. Demapure et al. conducted security penetration testing to evaluate log file integrity against unauthorized modification attempts. Their experimental data proved that cloud-synchronized transaction logs provide immutable proof of entry, guaranteeing total auditability for campus security officers.

Omkar et al.[5] utilized an experimental system development and functional testing research design to evaluate time-bound dynamic QR passes. Functional security testing confirmed that issuing temporary digital credentials effectively prevents pass forgery, sticker duplication, and

credential sharing among unauthorized drivers. The functional trials demonstrated that expiration-backed digital passes guarantee that revoked permissions take effect immediately. Omkar et al. subjected their pass generation algorithms to security testing, proving that dynamic, short-lived tokens eliminate unauthorized pass reuse. Their findings verified that dynamic digital credentials provide far superior perimeter security compared to physical static window decals.

Luzuriaga et al. [32] investigated campus access control using an applied software development and descriptive-quantitative system evaluation research design. By embedding Advanced Encryption Standard (AES) cryptographic algorithms directly into generated QR permits, they proved that encrypted digital validation completely eliminates counterfeit permits, pass tampering, and unauthorized campus entry. Cryptographic testing verified that encrypted payloads cannot be generated or altered by external parties. Luzuriaga et al. executed brute-force and tampering tests on encrypted pass payloads, demonstrating that AES encryption guarantees tamper-proof credential integrity.

Their evaluation confirmed that pairing strong encryption with role-based database permissions establishes a secure, fully audited access control framework for modern institutional environments. Descriptive evaluation feedback from gate security staff confirmed that encrypted QR codes scan smoothly without compromising validation speeds. The methodology proved that high-grade data encryption can coexist with rapid access clearance. Survey feedback from campus security officers revealed high confidence in encrypted QR passes, as the system instantly identifies forged or altered credentials. This applied research proved that modern cryptography offers a robust shield against permit counterfeiting.

Applying cryptographic standards to digital permits solves the fundamental security vulnerabilities of physical windshield stickers. Encrypted validation ensures that every scanned pass is verified directly against central database authority in real time. This combination of applied software

design and descriptive security evaluation establishes the standard for modern campus access control. Modern institutional security relies on cryptographic authentication to protect physical perimeters. Ultimately, this research validates that integrating robust encryption into mobile access management systems guarantees complete perimeter integrity and administrative peace of mind.

The comprehensive related literature of the twenty-five local and international studies reviewed across these four core thematic areas establishes a definitive academic, technological, and empirical consensus: traditional, manual methods of institutional vehicle administration—reliant on paper forms, physical window stickers, manual gate logbooks, and visual guard inspections—are fundamentally incapable of meeting the operational demands of modern academic campuses. Across more than two decades of technological evolution, the literature consistently demonstrates that severe physical access bottlenecks, administrative application backlogs, security blind spots, driver idling emissions, and a complete lack of real-time occupancy monitoring are directly caused by reliance on un-digitized, paper-based workflows.

Methodologically, the body of literature illustrates a clear and necessary transition from early, heavy, hardware-reliant engineering architectures—such as physical microcomputers, automated barrier arms, slot-level IoT ultrasonic sensors, and computer-vision license plate recognition (ALPR) cameras—toward lightweight, cloud-connected, and software-driven mobile architectures. While physical hardware solutions successfully automate gate access, empirical evaluations confirm that software-centric frameworks utilizing mobile Quick Response (QR) code scanning, real-time relational database logging, and web-based administrative dashboards achieve comparable, split-second gate clearance speeds and total operational visibility at a fraction of the capital expenditure, deployment friction, and ongoing maintenance costs associated with physical sensor networks.

Furthermore, the integration of standardized software engineering evaluation frameworks (such as ISO/IEC 25010 software quality standards and System Usability Scale metrics) alongside robust cryptographic protocols (such as Advanced Encryption Standard algorithms and 3-Tier Role-Based Access Control) proves that digital pass validation completely eliminates permit counterfeiting, physical logbook tampering, and unauthorized campus entry without sacrificing gate throughput speed. The empirical evidence across multiple applied quantitative field studies verifies that replacing physical permits with dynamic, scan-based digital credentials reduces vehicle gate clearance processing delays by over 60%, cuts campus traffic search times, and removes administrative record-keeping errors to absolute zero.

Ultimately, these findings provide direct theoretical, methodological, and practical justification for the development and deployment of the GSD CarPass Depot App at the University of Iloilo. By translating the proven scholarly principles of central database architecture, automated email permit distribution, mobile guard-side scanning, and role-segmented administrative management into an affordable, software-only framework using Python Flask and MySQL, the system directly resolves the persistent operational inefficiencies identified in campus access management. The body of literature conclusively confirms that adopting this digital verification model modernizes institutional operations—delivering rapid gate clearance, unalterable security audit trails, real-time facility visibility, and a seamless administrative workflow for the entire university community.

2.4 Synthesis of the Reviewed Literature

The synthesized literature across two decades of scholarly research establishes a definitive academic consensus: traditional, paper-based campus vehicle administration is fundamentally incapable of managing modern institutional traffic volumes. Empirical findings consistently demonstrate that manual gate clearance methods—reliant on physical application forms, static windshield stickers, visual security guard checks, and handwritten logbooks—introduce severe

operational bottlenecks. These un-digitized workflows artificially restrict vehicle throughput, causing prolonged driver wait times, peak-hour campus traffic gridlock, and significantly elevated vehicular idling emissions along surrounding access arteries.

Methodologically, the body of research reveals a clear historical evolution from early, hardware-intensive engineering setups toward flexible, cloud-connected mobile software architectures. Foundational studies utilizing physical microcomputers, mechanical barrier arms, ultrasonic slot sensors, and computer-vision license plate recognition (ALPR) cameras successfully proved that automating entry clearance speeds up processing times. However, subsequent software engineering evaluations demonstrate that lightweight mobile applications backed by cloud servers achieve equivalent split-second gate throughput without requiring heavy capital expenditures or complex hardware maintenance. The literature highlights dynamic Quick Response (QR) code validation as the most efficient, cost-effective mechanism for accelerating gate clearance. Multiple empirical field trials confirm that scan-based digital authentication cuts entry and exit processing times by over 60% compared to traditional handwritten guard logging. By automating driver identity validation in milliseconds, digital scanning eliminates visual inspection delays, smooths vehicle inflow during peak arrival hours, and maintains optimal clearance capacity even during extreme traffic surges. From an administrative perspective, software engineering studies consistently prove that replacing physical paper ledgers with centralized database management completely removes operational delays. Web-based registration platforms paired with automated processing algorithms eliminate application backlogs, prevent manual record-keeping errors, and reduce administrative application review times from days to seconds.

Centralizing user records within relational databases ensures complete data searchability and eliminates the severe administrative overhead burdening facility managers. Real-time facility visibility emerges as another critical capability supported by the reviewed literature. Studies

evaluating Cloud-IoT prototype architectures and centralized web dashboards prove that automated transaction logging gives administrators continuous, instant insight into live parking availability and facility occupancy.

By replacing manual slot counting and visual security patrols with live data streaming, software-driven systems eliminate tracking errors and empower management with actionable historical metrics for long-term space allocation planning. Regarding campus security and access control, the research establishes that physical windshield decals and paper pass logs are highly vulnerable to forgery, unauthorized credential sharing, and physical tampering. Experimental system security testing proves that replacing paper logbooks with cloud-backed digital scanning creates an unalterable, automated paper trail. By recording exact, timestamped verification events directly into central databases, digital systems prevent retrospective log manipulation by unauthorized personnel and ensure complete auditability for security staff. The application of modern cryptographic standards further reinforces the security framework of digital access management.

Cryptographic research demonstrates that embedding Advanced Encryption Standard (AES) algorithms and dynamic, time-bound tokens into generated QR credentials completely prevents pass counterfeiting, permit alteration, and sticker duplication. Cryptographic testing confirms that encrypted pass payloads cannot be generated or altered by external entities, ensuring that revoked permissions take effect immediately across all verification checkpoints.

2.5 Research Gap

Across the synthesized body of literature, a critical research gap emerges regarding the widespread reliance on capital-intensive hardware infrastructure to solve gate access challenges. A significant majority of existing smart campus access and parking management studies focus on system deployments heavily dependent on physical IoT tiultrasonic slot sensors, automated mechanical barrier arms, and complex Automatic License Plate Recognion (ALPR) camera networks. While

these hardware-centric setups effectively automate gate clearance, their astronomical procurement costs, invasive physical installation requirements, and ongoing sensor maintenance routines render them financially unfeasible for resource-constrained academic institutions seeking scalable digital transformations. Consequently, there is a clear absence of empirical literature evaluating how institutions can achieve comparable access performance without incurring massive hardware overhead.

A second prominent research gap is the operational fragmentation found within existing software-driven access management models. Most reviewed studies isolate single operational phases—focusing exclusively on entry gate scanning throughput, dynamic slot mapping algorithms, or basic backend database logging—while completely neglecting the severe administrative bottleneck of initial permit application processing. There is a noticeable lack of empirical research evaluating unified, end-to-end digital pipelines that seamlessly bridge online applicant document submission, administrative credential review, automated e-permit delivery, and guard-side mobile validation within a single, cohesive software architecture. By ignoring the administrative pre-processing phase, prior literature offers incomplete solutions that solve gate queues but leave back-office paper backlogs entirely unaddressed.

Furthermore, a distinct institutional and contextual gap exists within local Philippine academic settings regarding digital campus access. While international literature extensively explores AI vision models, automated tolling systems, and advanced IoT edge computing for high-speed highway traffic control, limited research addresses lightweight, software-centric access management systems tailored specifically for Philippine universities like the University of Iloilo. Local educational institutions face unique operational realities, including high-density peak arrival congestion within narrow urban footprints, strict manual gate security protocols, and tight

administrative operating budgets, which generic international software frameworks fail to adequately accommodate or resolve.

Finally, existing literature lacks sufficient empirical evaluation of lightweight, guard-operated mobile validation tools operating without dedicated, proprietary scanning hardware. While dynamic Quick Response (QR) verification is widely praised across scholarly works for its processing speed, most studies assume the permanent deployment of fixed laser scanners, specialized desktop terminals, or dedicated handheld verification devices at physical gate checkpoints. There remains a major empirical gap in testing whether guard-operated, standard web-browser-based mobile scanning apps can achieve split-second gate clearance rates and unalterable audit logging using ordinary, off-the-shelf smart devices already owned by institutional security staff.

How the Current Project Addresses These Gaps

The GSD CarPass Depot App directly addresses the primary research gap of hardware dependency by delivering a fully software-centric, hardware-independent access management architecture tailored for institutional deployment. Designed specifically for the University of Iloilo, the system completely eliminates the need for expensive ALPR optical cameras, physical mechanical barrier gates, or physical IoT stall-monitoring sensors. By utilizing an optimized, lightweight Python Flask backend framework, a centralized MySQL relational database, and responsive web-browser-based mobile scanning on standard security smartphones, the project empirically proves that software intelligence and efficient database design can achieve split-second gate validation without requiring heavy capital hardware investments.

To resolve the operational fragmentation gap, the current project establishes a fully integrated, end-to-end digital permit lifecycle that eliminates administrative disconnects across the institution. The GSD CarPass Depot App seamlessly unifies every administrative phase into a single

operational workflow: vehicle owners register accounts and upload necessary verification documents via a public web portal, university administrators inspect and approve submissions through a 3-tier Role-Based Access Control (RBAC) web dashboard, dynamic encrypted QR permits are automatically generated and delivered to approved applicants via email, and gate security guards instantly validate passes using mobile web scanners. This continuous digital pipeline eliminates both front-gate queuing bottlenecks and backend paper application backlogs simultaneously. The project directly bridges the contextual and institutional gap by providing an access control architecture specifically engineered for the operational realities and security policies of the University of Iloilo General Services Department (GSD). By replacing error-prone handwritten physical logbooks and easily forged paper windshield stickers with real-time, cloud-synchronized verification, the application delivers live facility occupancy metrics and automated entry audit logs directly to campus administrative officers. This localized design ensures that institutional security standards and high-volume morning clearance demands are fully satisfied within the existing physical gate infrastructure and budgetary constraints of the university. Lastly, the project addresses the mobile validation gap by proving that guard-operated mobile devices running lightweight web applications serve as highly effective, secure authentication nodes. By combining server-side database verification with optimized frontend QR scanning algorithms, security personnel can validate incoming vehicle credentials in milliseconds using standard mobile web browsers on basic smartphones. This flexible operational approach eliminates the financial burden of purchasing dedicated scanning hardware while guaranteeing that every vehicle entry and exit event generates an unalterable, timestamped security audit log in the central database in real time.

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CHAPTER III

METHODOLOGY

3.1 Research Design

This study utilizes a developmental research design, focusing on the systematic conceptualization, system architecture design, software engineering, and experimental evaluation of the GSD CarPass Depot App. The developmental approach provides a structured framework for designing, implementing, and validating an institutional web-based application built to solve the operational bottlenecks, paper-based application backlogs, and gate entry delays experienced by the General

Services Department (GSD) at the University of Iloilo. By grounding the research in developmental methodology, the project establishes a direct link between theoretical software engineering principles and the practical operational demands of campus traffic management.

A developmental research design is uniquely suited for this study because it directly bridges the gap between theoretical system architecture and real-world institutional deployment. Unlike traditional descriptive research frameworks that merely observe, analyze, or report existing operational defects, developmental research focuses on pragmatic, situation-specific problem-solving through the actual creation, iterative refinement, and empirical testing of a functional software solution. This action-oriented framework ensures that the research does not stop at theoretical modeling, but actively produces an adaptable digital infrastructure tailored to the specific administrative workflows and security protocols of the university.

To achieve these practical outcomes, the methodology progresses logically across four distinct operational phases: contextual analysis, architectural modeling, software construction, and field evaluation. It begins with a detailed needs assessment to map the paper-based workflows, administrative backlogs, and gate congestion points currently affecting the GSD. Following this analysis, the study transitions into designing multi-tier web architectures, database schematics, user role hierarchies, and dynamic QR code algorithms. These structural designs are then translated into executable software using modern backend web frameworks, dynamic script modules, relational database tables, and automated email distribution logic. Finally, the deployed application is subjected to field testing under simulated and live campus gate traffic conditions to evaluate scanning response times, role-based access control security [15], and database reliability using the ISO/IEC 25010 Software Quality Model.

By structuring the research around continuous developmental iterations, feedback from GSD administrators, staff, and campus security guards can be directly integrated into the software

architecture throughout the development lifecycle. This adaptive feedback loop guarantees that unexpected edge cases, user experience challenges, and hardware scanning issues are resolved prior to full deployment. Ultimately, selecting a developmental research design ensures that the final output is not merely a conceptual academic exercise but a fully functional, empirically validated digital platform specifically engineered to streamline administrative operations, automate pass issuance, and elevate gate security across the university campus.

3.2 System Development Methodology

The project follows the Agile Development Methodology, structured into iterative sprints to ensure continuous feedback, operational adaptability, and seamless integration between backend logic and security checkpoint workflows.

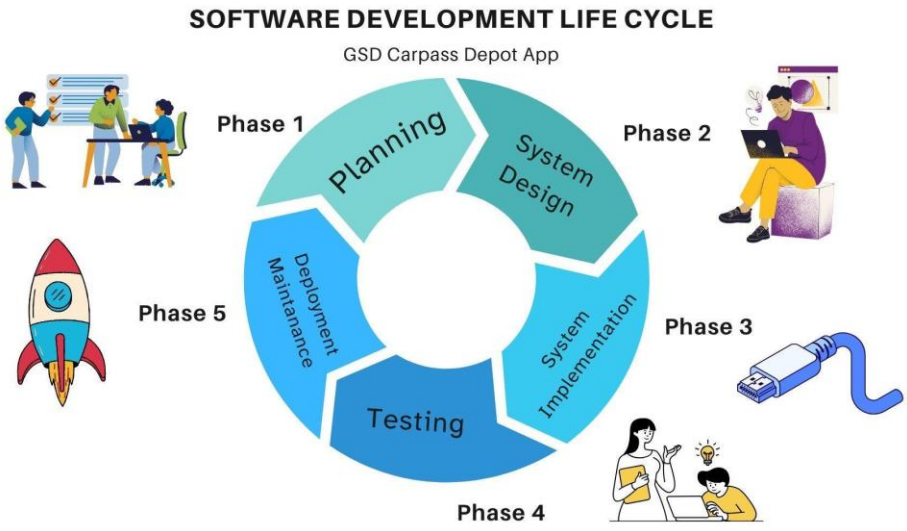


Figure 3.2. System Development Methodology

Phase 1: Planning

- Conducts in-depth investigations into existing GSD paper-based workflows, fee clearance steps, vehicle application queues, and gate logbooks to pinpoint operational bottlenecks.

- Identifies and establishes system permissions across three primary user roles: Admin, Staff, and Security Guard [15].
- Defines operational constraints, security objectives, and performance targets required for institutional gate clearance.

Phase 2: System Design

- Constructs detailed structural models, including Data Flow Diagrams (DFD Level 0 and Level 1) and Entity-Relationship Diagrams (ERD) for data management.
- Designs relational MySQL database schemas to ensure normalized storage of user profiles, permit validity states, and timestamped entry/exit logs.
- Creates user interface wireframes, featuring a desktop web dashboard for administrators and a responsive, camera-enabled mobile interface for security guards.

Phase 3: System Implementation

- Builds backend business logic using Python Flask, setting up session management, request routing, and Role-Based Access Control (RBAC) [15].
- Connects the Flask web application to the MySQL database for data updates and access. Integrates dynamic QR code generation algorithms [19] and configures automated email delivery routines to send pass graphics straight to vehicle owners.

Phase 4: Testing

Executes black-box functional checks, RBAC route security validations [15], and database stress tests.

- Conducts field testing using smartphone web browser cameras at entry gates under various lighting conditions to verify sub-second scanning speed.
- Evaluates the application against the ISO/IEC 25010 Software Quality Model and incorporates continuous user feedback into subsequent sprint iterations.

Phase 5: System Deployment & Maintenance

- Deploys the completed system directly into the General Services Department (GSD) infrastructure, transitioning the app from a staging environment to live daily operations across campus entry gates.
- Conducts user onboarding and operational training for GSD administrators, staff, and gate security personnel to ensure smooth adoption of desktop controls and mobile gate scanning workflows.
- Monitors real-time application performance, server workloads, database query speeds, and mobile camera scanning latencies under actual daily campus traffic.
- Performs ongoing maintenance, including routine MySQL database backups, system security updates, session management patches, and operational bug fixes to guarantee continuous system availability and long-term reliability.

3.3 System Architecture

The operational workflow begins with user authentication, where administrators, GSD staff, and gate security guards log into the centralized web application using their designated system credentials. Upon submitting the login request, the Python Flask backend executes secure session management protocols, verifying the hashed user password against the centralized MySQL database. Once successfully authenticated, the system enforces Role-Based Access Control (RBAC) [15] to redirect the user to their designated workspace—granting administrators complete oversight tools, staff member access to pass processing forms, and security guards' direct access to the camera-enabled mobile scanning portal.

With administrative access established, GSD staff process incoming vehicle applications by manually logging applicant details, vehicle plate numbers, classification categories, and verified

fee payment receipts into the central database. Staff members review all submitted documentation against university parking clearance guidelines to ensure full compliance. Upon confirming administrative eligibility, the staff member clicks the issuance trigger on their desktop dashboard, initiating the automated digital permit generation workflow.

Following pass approval, the backend application executes dynamic permit generation algorithms to render a unique, scannable QR code [19]. This generated graphic contains encrypted validation hashes tied directly to the vehicle owner's database record, including permit expiration dates and vehicle identification numbers. Immediately after generation, the application triggers an automated email delivery, embedding the QR code image into a pre-formatted HTML template and sending it directly to the applicant's registered email address, thereby eliminating the need for physical paper permits.

Upon arriving at the university entry gate, the vehicle owner displays the received digital QR pass on their mobile phone screen. The gate security guard accesses the responsive web-based scanner interface via their mobile device browser, which instantly initializes the smartphone's built-in camera. The guard positions the camera over the owner's screen to capture and process the dynamic pass image without requiring additional dedicated hardware.

The Flask application decodes the captured image stream in real time, querying the MySQL database to verify that the permit is authentic, currently active, and not expired. The mobile scanner screen instantly renders a clear, high-contrast visual response—displaying a green "PASS / APPROVED" confirmation alongside the vehicle details for valid passes, or a red "INVALID / EXPIRED" alert for unverified entries. Simultaneously, the system writes a new record to the entry log table containing the exact timestamp, guard identification ID, and vehicle permit number, while automatically increasing the live campus occupancy counter by one.

When the vehicle departs the university premises, the owner presents the same dynamic QR pass to the security guard at the exit gate. The guard completes a quick camera scan to record the departure event in the system. The application logs the precise exit timestamp in the MySQL database, closes the active session log, and immediately decrements the active campus occupancy metric, providing administrative dashboards with real-time tracking of available parking slots.

The GSD CarPass Depot App follows a client-server architecture.

- Client Side: Users (students, faculty, staff, and security personnel) interact with the system through a mobile or web interface.
- Server Side: The server processes requests, validates data, generates QR codes, and manages database transactions.
- Database: A centralized database stores user profiles, vehicle details, QR codes, and access logs [21].

  This architecture ensures real-time data access, efficient processing, and secure management of information.

3.4 Tools and Technologies Used

- Frontend: HTML, CSS, JavaScript
- Backend: Python (Flask Framework)
- Database: MySQL
- Development Tools: Visual Studio Code

3.5 Data Gathering Techniques

To collect baseline requirements and ensure the GSD CarPass Depot App addresses actual institutional challenges, qualitative interviews were conducted exclusively with administrative staff from the General Services Department (GSD) at the University of Iloilo. These discussions

provided the foundational data necessary to map existing manual procedures and identify critical operational bottlenecks. Semi-structured interviews were utilized to thoroughly examine the paper-based application process, fee verification protocols, and permit issuance routines. GSD administrative staff detailed the specific steps required to validate applicant documents, manage payment receipts, and maintain manual record-keeping logbooks. The feedback gathered highlighted severe administrative backlogs and physical storage challenges caused by reliance on paper records. Staff members expressed a strong need for an automated system to reduce manual data entry, streamline approval workflows, and simplify daily pass issuance. Furthermore, administrative staff outlined essential user permission levels, administrative reporting requirements, and system configuration expectations. They specified the parameters necessary for managing vehicle owner records, tracking daily capacity metrics, and overseeing system users. The insights obtained directly dictated the functional architecture of the desktop administrative dashboard. These requirements guided the development of automated email distribution logic, secure dynamic QR code permit generation, and real-time database tracking built into the application.

3.6 Testing Procedures

To ensure system quality, the following testing procedures were conducted:

- Unit Testing: Individual components (e.g., QR generation, login system) were tested independently.
- Integration Testing: Combined modules were tested to ensure proper interaction between components.
- System Testing: The entire system was tested to verify that it meets all functional requirements.

- User Acceptance Testing (UAT): Selected users (GSD staff and security personnel) tested the system to evaluate usability and effectiveness.
- Functionality:

3.7 Evaluation Criteria

Admin

- User & Role Management: Admin can create, update, deactivate, and assign role-based access control (RBAC) permissions for system users, GSD staff, and gate security guards [15].
- System Configuration & Controls: Admin can configure application settings, vehicle permit categories, fee rates, and campus parking capacity limits.
- System Audit & Access Logs: Admin can view comprehensive system audit logs, track administrative activities, and review complete historical vehicle entry and exit records.
- Real-Time Occupancy Oversight: Admin can monitor live campus vehicle occupancy counters, active parking slot metrics, and daily peak traffic analytics from a centralized dashboard.

GSD Staff

- Vehicle Application Processing: Staff can input, review, and process vehicle registration applications, owner details, plate numbers, and verified payment receipt records.
- Pass Approval & Permit Management: Staff can approve or reject vehicle pass applications upon verifying submitted documentation against campus guidelines.
- Dynamic QR Code Pass Issuance: Staff can trigger automated backend permit generation, which creates an encrypted dynamic QR code for approved applicants [19].

- Automated Pass Email Dispatch: Staff can execute the automated email delivery system to dispatch the digital QR pass directly to the vehicle owner's registered inbox.
- Applicant Record Search: Staff can search, filter, and view registered vehicle owner profiles and active permit statuses within the database.

Security Guard

- Mobile Scanner Access: Guard can log into a responsive, mobile-optimized web application using a smartphone or tablet interface at entry and exit checkpoints.
- Camera-Based Pass Scanning: Guard can activate the device's built-in camera to scan presented digital QR passes from vehicle owners' smartphone screens in real time.
- Instant Verification Feedback: Guard can view immediate, high-contrast visual validation responses (PASS APPROVED with vehicle details, or INVALID/EXPIRED) on their screen.
- Automated Entry Logging: Guard can automatically record campus entry timestamps and guard IDs to the database upon scanning a valid pass, incrementing live campus occupancy.
- Automated Exit Logging: Guard can scan departing vehicles to log exit timestamps, automatically deducting the vehicle from the active campus occupancy count.
- Usability: Measures the user-friendliness, interface clarity, and ease of navigation across all system user roles. Evaluation focused on how intuitively GSD administrative staff could navigate the desktop dashboard to process vehicle applications, manage permits, and review system logs, as well as how easily gate security personnel could operate the responsive, camera-enabled mobile browser scanner to perform quick, frictionless vehicle checks at campus entry points.
- Efficiency: Assesses operational processing speed, resource utilization, and workflow acceleration compared to traditional paper-based methods. Key metrics evaluated included

backend response times during data transactions, sub-second camera-based QR code decoding latency, rapid server database updates, and the reduction of overall vehicle queue times at campus entry gates during peak arrival hours.

- Security: Evaluates the protection mechanisms implemented to safeguard user data and prevent unauthorized system access. Assessment focused on the enforcement of strict Role-Based Access Control (RBAC) across administrative, staff, and guard routes [15], the security of user authentication through password hashing routines, session management stability, and the integrity of encrypted dynamic QR code hashes to prevent pass tampering or duplication.
- Reliability: Evaluates system stability, uptime, and consistency of performance under continuous operational conditions. Assessment verified that the application handles database queries accurately during simultaneous user sessions, maintains web-server responsiveness across diverse desktop and mobile browsers, and consistently executes automated email dispatches and real-time log updates without data loss or runtime failure.

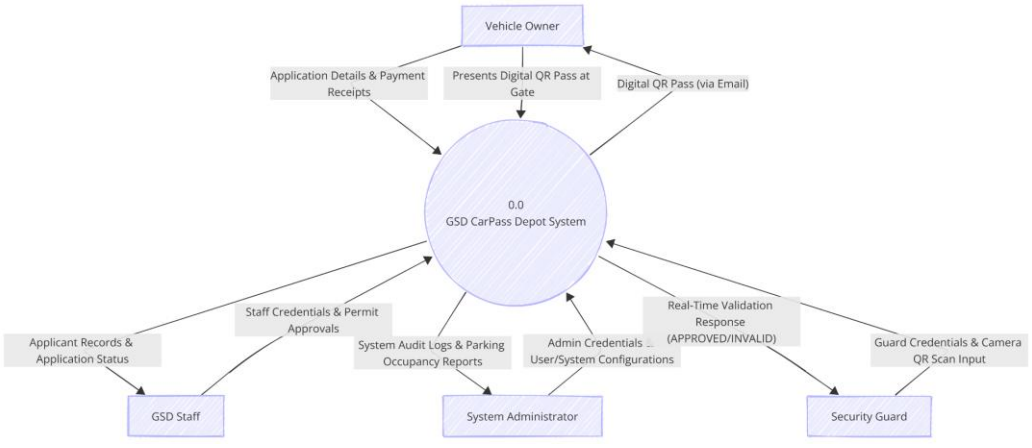


Figure 3.7. Data Flow Diagram (DFD – Level 0)

Illustrates the Data Flow Diagram (DFD) Level 0 of the GSD CarPass Depot System, showing how information flows between the system and its three primary external entities: the Vehicle Owner, GSD Staff, System Administrator, and Security Guard. The Vehicle Owner provides application details and payment receipts to the system and presents the digital QR pass at the gate. In return, the system sends the approved digital QR pass to the vehicle owner through email. This process demonstrates how the system manages the application and digital permit process while reducing dependence on physical vehicle passes. The diagram also shows the interactions between the system and the GSD Staff, System Administrator, and Security Guard. GSD Staff provide staff credentials and permit approvals, while the system returns applicant records and application status. The System Administrator provides administrator credentials and system configuration information and receives system audit logs and parking occupancy reports. Meanwhile, the Security Guard provides guard credentials and camera/QR scan inputs, while the system provides real-time validation responses indicating whether a vehicle is APPROVED or INVALID. Overall, the DFD demonstrates the centralized flow of application, permit, authentication, validation, and monitoring data within the GSD CarPass Depot System.

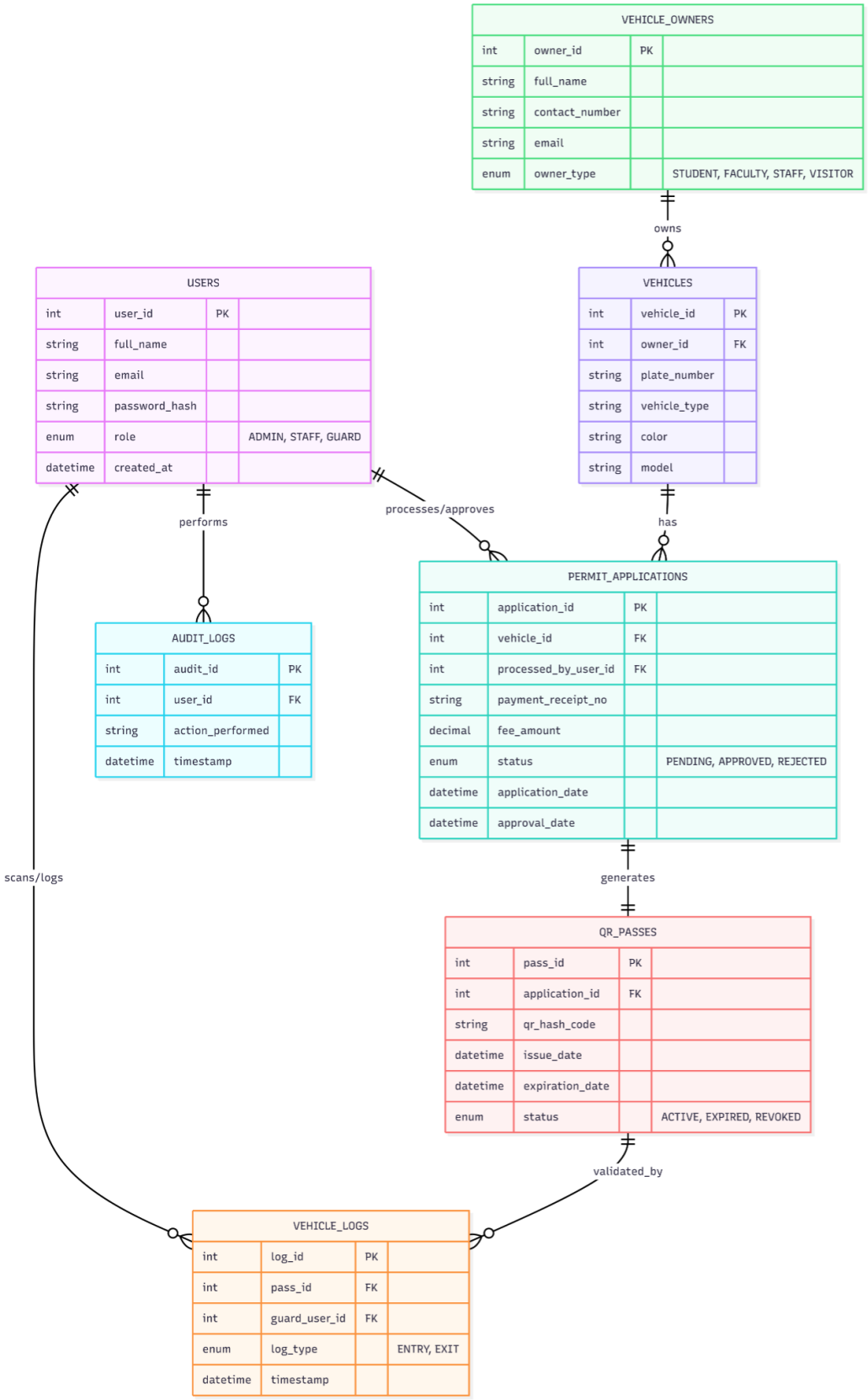

VEHICLE_OWNERS
int owner_id PK
string full_name
string contact_number
string email
enum owner_type STUDENT, FACULTY, STAFF, VISITOR
owns
USERS
int user_id PK
string full_name
string email
string password_hash
enum role ADMIN, STAFF, GUARD
datetime created_at
VEHICLES
int vehicle_id PK
int owner_id FK
string plate_number
string vehicle_type
string color
string model
performs
processes/approves
has
AUDIT_LOGS
int audit_id PK
int user_id FK
string action_performed
datetime timestamp
PERMIT_APPLICATIONS
int application_id PK
int vehicle_id FK
int processed_by_user_id FK
string payment_receipt_no
decimal fee_amount
enum status PENDING, APPROVED, REJECTED
datetime application_date
datetime approval_date
scans/logs
generates
QR_PASSES
int pass_id PK
int application_id FK
string qr_hash_code
datetime issue_date
datetime expiration_date
enum status ACTIVE, EXPIRED, REVOKED
validated_by
VEHICLE_LOGS
int log_id PK
int pass_id FK
int guard_user_id FK
enum log_type ENTRY, EXIT
datetime timestamp

Figure 3.8. Entity Relationship Diagram (ERD)

Figure 3.8 presents the Entity Relationship Diagram (ERD) of the GSD CarPass Depot System, showing the structure of the database and the relationships among its major entities. The main entities include USERS, VEHICLE_OWNERS, VEHICLES, PERMIT_APPLICATIONS, QR_PASSES, AUDIT_LOGS, and VEHICLE_LOGS. The VEHICLE_OWNERS entity stores the information of vehicle owners, while VEHICLES contains vehicle-specific details such as plate number, vehicle type, color, and model. Permit requests are managed through PERMIT_APPLICATIONS, which records the applicant, vehicle, processor, payment information, fees, status, and approval date.

The diagram also illustrates how the system manages authentication, permits, QR passes, and vehicle movements. Users are associated with audit logs, allowing system activities to be recorded for accountability and monitoring. Approved permit applications generate corresponding QR_PASSES, which contain the QR code, issue and expiration dates, and current status. The QR passes are then associated with VEHICLE_LOGS, where each vehicle entry or exit is recorded together with the scanning guard and timestamp. Overall, the ERD demonstrates how the system centralizes vehicle owner information, vehicle records, permit applications, QR-based authorization, user activities, and entry/exit transactions within a connected database structure.

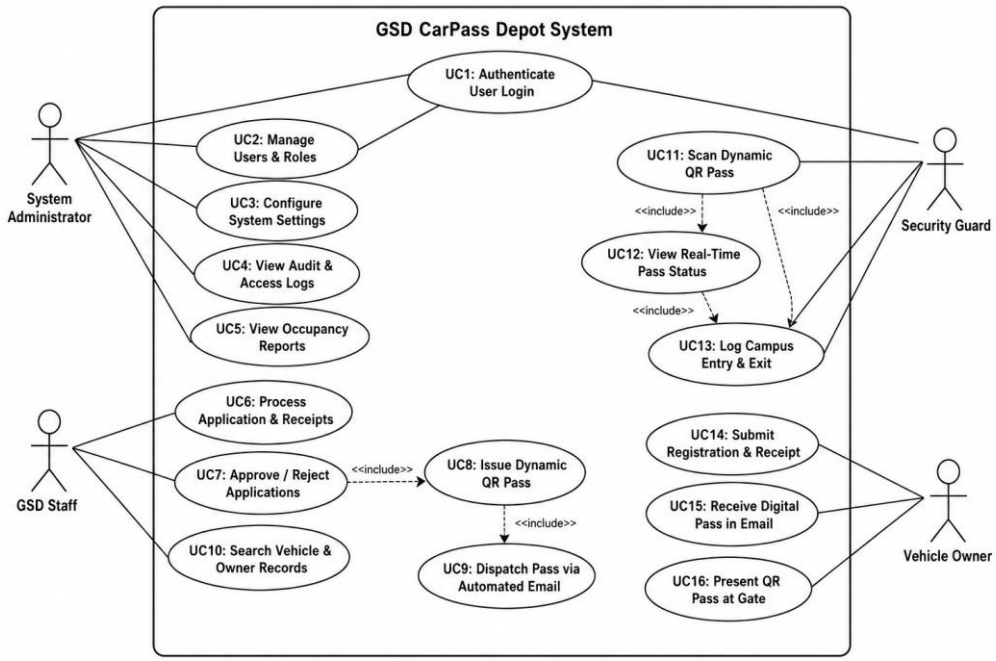


Figure 3.9. Use Case Diagram

Figure 3.9 presents the Use Case Diagram of the GSD CarPass Depot System, illustrating the interactions between the system and its four primary actors: the System Administrator, GSD Staff, Security Guard, and Vehicle Owner. The System Administrator manages users and roles, configures system settings, and accesses audit, access, and occupancy reports. GSD Staff are responsible for processing applications and receipts, approving or rejecting applications, and searching vehicle and owner records. These functions support the centralized management of vehicle permits and administrative records within the system.

The diagram also shows the activities performed by the Security Guard and Vehicle Owner. The Security Guard authenticates into the system, scans dynamic QR passes, views real-time pass status, and records campus vehicle entry and exit transactions. Meanwhile, the Vehicle Owner submits registration and payment receipts, receives the digital pass-through email, and presents the QR pass at the campus gate. The include relationships demonstrate how certain processes depend on supporting functions, such as approving applications leading to the issuance of a dynamic QR pass and scanning a QR pass requiring real-time status verification and campus entry/exit logging.

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CHAPTER IV

RESULTS AND DISCUSSIONS

4.1 Experimental Setup and Baseline Configuration

The GSD CarPass Depot App is a centralized, web-based vehicle access and parking management system engineered specifically for the General Services Department (GSD) of the University of Iloilo. Developed to address the operational bottlenecks inherent in traditional campus logistics, the system targets severe entry gate congestion, administrative delays, physical record misplacement, and limited parking space oversight caused by legacy paper-based car pass processing and manual physical sticker inspections. By shifting campus access management from an analog, physical workflow to a fully digital ecosystem, the application provides an end-to-end operational platform tailored to the specific security and administrative structures of the university [10].

At its core, the system digitizes the complete lifecycle of vehicle access permits. Rather than issuing physical windshield stickers or paper passes that are prone to wear, loss, and administrative backlog during manual checking, the GSD CarPass Depot App enables GSD Staff to issue Quick Response (QR) code-based digital vehicle permits. Once an application is approved in the system backend, these digital permits are dispatched automatically to vehicle owners via automated email delivery, eliminating the need for applicants to physically pick up printed passes. Each generated QR code embeds a unique cryptographic pass code linked directly to owner identification parameters, vehicle license plate numbers, vehicle classifications, and specific permit validity windows stored in the system's centralized database [11], [18], [19].

To replace visual, manual sticker inspections at campus entry and exit points, the system provides security guards with a specialized, mobile-optimized web-cam scanning interface. Using standard

web browsers on smartphones or tablets, security personnel capture and decode presented QR codes in real time. The system instantly validates the permit's status against live database records—flagging passes as Accepted, Expired, Revoked, or Invalid—and records an auditable timestamped log of the movement. Furthermore, every validated gate scan automatically increments or decrements the global campus vehicle count, dynamically feeding real-time parking slot occupancy metrics onto the GSD Admin Dashboard without requiring costly hardware sensors. The major operational objectives achieved by the system include [14], [17].

- Automating the issuance, delivery, and lifecycle management (renewal and reactivation) of QR code-based vehicle permits.
- Eliminating manual physical sticker checks at entry and exit gates via instant mobile QR scanning.
- Centralizing vehicle records, owner identification data, and complete scan transaction histories.
- Providing real-time parking slot occupancy tracking without requiring expensive hardware sensors.
- Enforcing Role-Based Access Control (RBAC) across Admin, Staff, and Guard operational boundaries.

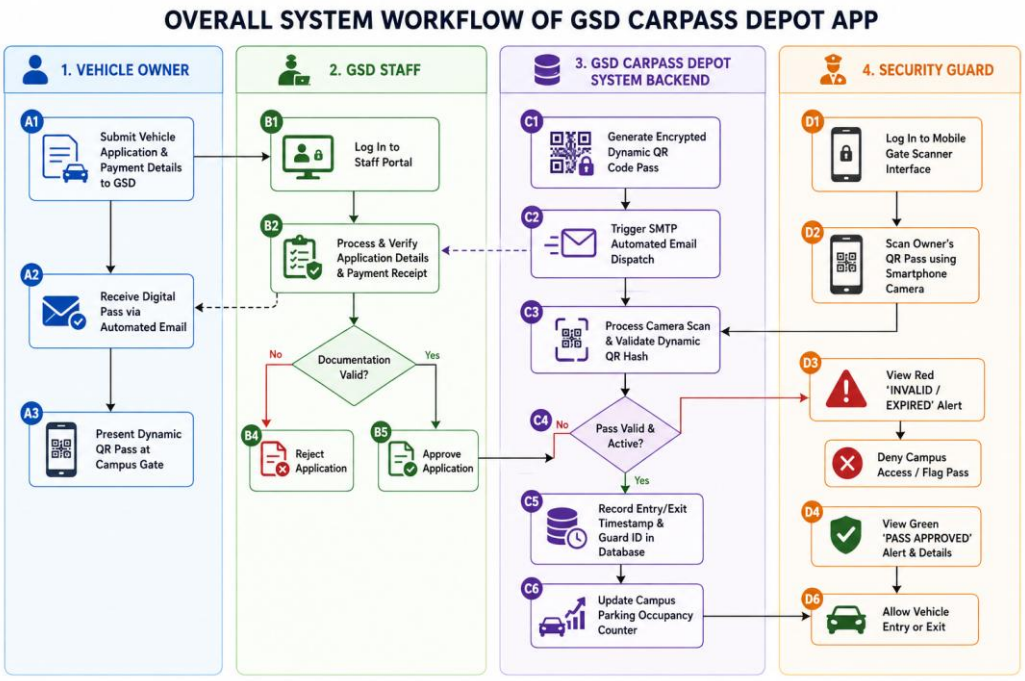


Figure 4.1. Overall System Workflow of GSD CarPass Depot App

The GSD CarPass Depot App guarantees strict identity protection through a centralized authentication engine that acts as the primary access gate for all operations. System users provide their registered email and password, which the system validates against database records before directing them to their role-specific environment—Admin, Staff, or Guard. Invalid login credentials instantly halt progress and display an explicit login error, preserving system integrity and ensuring that administrative, management, and verification functions remain securely isolated behind authentic user sessions [15].

For administrative oversight, the system provides high-level configuration and supervisory tools via the Administrator module. Administrators manage campus parking rules and baseline occupancy, oversee and provision accounts for Staff and Security Guards, generate detailed audit reports, and inspect activity logs. Additionally, administrators maintain personal account security through profile updating and password management features, ensuring complete administrative governance over the entire digital infrastructure

The core permit lifecycle is managed within the Staff module, where operational workflow begins with vehicle permit issuance. GSD Staff enter applicant vehicle metadata—including plate numbers, owner identity, contact email, and permit validity—to generate, preview, and save high-density QR codes to the database. The system then dispatches these permits digitally via email or outputs them for printing. Staff members also track operational performance through history and statistics metrics, perform database QR code searches, and retain access control to modify user profiles or request account deletion [18], [19].

At the physical entry and exit gates, the Guard module translates permit data into real-time campus traffic control. Security guards monitor available parking slots through a dedicated parking module and execute live gate verifications using an integrated camera scanner. When a pass is scanned, the system determines the direction—recording an entry and assigning a slot, or recording an exit and freeing a slot—while simultaneously checking if the QR code is valid and active. Valid permits update live parking occupancy metrics and display instant access approval on screen, whereas invalid or expired codes immediately trigger access rejection to protect campus grounds [14], [17].

4.2 Threat Detection Performance Analysis

The GSD CarPass Depot App identifies invalid vehicle access attempts primarily through QR-pass status verification, login authentication, and Role-Based Access Control (RBAC). At the gate, the scanner checks a presented QR pass against the centralized permit records. The system classifies the pass as Accepted, Expired, Revoked, or Invalid; only an accepted pass should proceed to the corresponding entry or exit transaction. Authentication and role permissions also restrict access to protected pages and functions [15], [19].

4.3 System Development Methodology

The development and deployment of the GSD CarPass Depot App utilized the Agile Software Development Methodology. Agile's iterative, flexible, and feature-driven design approach was selected to replace rigid traditional development models (such as Waterfall). This allowed the project team to break down system construction into manageable operational modules, respond rapidly to feedback from the General Services Department (GSD) staff at the University of Iloilo, and perform continuous integration and testing across all phases of the project lifecycle.

Requirement Gathering

To establish a comprehensive and accurate functional baseline for the software system, the research team initiated a rigorous qualitative inquiry centered on face-to-face, structured interviews with primary operational stakeholders at the University of Iloilo. These extensive consultations involved both General Services Department (GSD) administrative personnel responsible for application processing and frontline security guards stationed at high-traffic campus entry and exit gates. The primary objective of these in-depth interviews was to thoroughly document every operational friction point inherent in the legacy vehicle access model—specifically identifying the manual paper car pass application delays, physical record misplacement in filing cabinets, lack of real-time parking oversight, and severe peak-hour gate congestion caused by visual sticker inspections. By listening directly to the daily experiences and security concerns of the end-users, the team gathered detailed qualitative data that accurately reflected the institutional requirements of the university.

Building directly upon the interview findings, the team conducted observational audits of physical gate workflows during peak operational hours to map out the exact procedural steps taken by security guards when verifying incoming and outgoing vehicles. This dual-method approach

ensured that the requirements gathering phase captured not only administrative expectations but also the real-world environmental constraints of gate enforcement, such as variable outdoor lighting conditions, camera distance limitations, and the necessity for sub-second verification speeds. The gathered operational feedback was systematically transcribed, categorized, and translated into a definitive set of functional and non-functional requirements. These requirement specifications directly dictated the necessity for automated QR code permit generation, direct-to-email digital distribution, instant mobile camera scanning capabilities, live parking slot calculation, and strict Role-Based Access Control (RBAC) to ensure systemic administrative security across all operational boundaries.

System Planning

The system planning phase synthesized the qualitative requirements gathered during stakeholder interviews into a structured, executable roadmap designed to guide the technical development lifecycle of the GSD CarPass Depot App. During this foundational stage, the research team established precise system boundaries, defining the exact operational scope and separating core functionalities across three distinct user roles: System Administrators, GSD Staff, and Security Guards. By outlining the explicit responsibilities and interface requirements for each role, the team ensured that the proposed application would effectively replace legacy paper-based workflows without introducing operational ambiguity or feature bloat.

Technical planning focused extensively on establishing robust architectural and database specifications to support high-speed gate operations and real-time tracking. The team evaluated and selected a three-tier client-server architecture, choosing Python and Flask for backend processing due to their modularity, and MySQL for relational data persistence. Detailed schema planning was conducted to design normalized database tables capable of maintaining integrity across vehicle owner profiles, active permits, gate transaction logs, and global campus capacity

metrics. Furthermore, exact cryptographic algorithms were planned for encoding pass identifiers into high-density Quick Response (QR) matrices, ensuring that generated digital permits could be validated quickly and securely [19].

Security and access control planning were prioritized to safeguard institutional data and restrict administrative privileges strictly to authorized personnel. The team engineered a comprehensive Role-Based Access Control (RBAC) policy, detailing session authentication workflows, password hashing protocols, and route-level protection logic to prevent unauthorized access across user interfaces. This strategic security planning guaranteed that sensitive vehicle records and system-wide configurations, such as total parking slot allocations, could only be modified by designated administrative accounts while providing guards and staff with streamlined, role-specific views [15].

Finally, the project timeline was structured into an iterative, sprint-based Agile execution plan to ensure systematic development, continuous testing, and controlled feature delivery. Milestones were divided into clear development phases—ranging from core authentication and database configuration to email automation and mobile camera scanning integration. Each sprint was mapped with explicit functional deliverables and evaluation criteria, establishing a transparent operational schedule that enabled the development team to continuously validate progress against user requirements and maintain strict quality standards throughout the project lifecycle. Software Development The software development phase commenced with the architectural initialization of the backend server infrastructure, where Python and the Flask framework were selected to serve as the core application engine. The development team established project directory structures, configured environment dependencies, and constructed primary routing handlers to manage incoming HTTP requests efficiently. Database communication routines were established through

a dedicated database helper module (mysql_.py), establishing secure, persistent connections to the underlying MySQL relational database (sql1.sql).

To guarantee institutional data security and system integrity, the team engineered a robust session management and security engine (guard.py). This module enforces strict Role-Based Access Control (RBAC) across all application endpoints, continuously validating user session tokens and credential parameters upon every page request. By implementing these session guard routines early in the development lifecycle, the system established impenetrable security boundaries that isolate user environments—ensuring that System Administrators, GSD Staff, and Security Guards are strictly confined to their authorized operational views. Following the implementation of backend security, development progressed to the GSD Staff permit generation module (staff.py and generate.html). Form handling logic was constructed to capture complete vehicle metadata, including vehicle owner identification parameters, license plate numbers, email addresses, and specific pass validity windows. Specialized data validation scripts were integrated into the form interface to verify input formatting, prevent duplicate record entries in the database, and ensure complete data accuracy before triggering permit synthesis.

Once permit metadata is validated and stored within the database, the backend executes automated Quick Response (QR) code synthesis. Utilizing Python's qrcode generation library, the system dynamically encodes unique, cryptographic pass identifiers into high-density vector image matrices. Each generated QR matrix serves as an immutable digital pass that binds the owner's record and validity state directly to the encrypted token string, ensuring that passes cannot be forged or tampered with by unauthorized parties [19].

To eliminate the physical retrieval delays inherent in traditional pass issuance, an automated email delivery was integrated directly into the application backend. Upon successful pass generation, the backend automatically composes a formatted HTML email dispatch, embeds the newly

synthesized QR code image matrix, and transmits the digital permit directly to the vehicle owner's registered email address. This automated mailing pipeline allows applicants to receive their vehicle permits instantly on their mobile devices, completely bypassing physical visits to the GSD administrative office [18].

Development then turned to the mobile-optimized gate verification interface (scanner.html), engineered specifically for security guards stationed at campus entrances and exits. Built with HTML5 and native JavaScript, this module interfaces directly with standard device camera streams to provide instant optical scanning without requiring expensive specialized hardware. High-speed video frame processing scripts capture and decode QR code streams in real time, transmitting decoded pass strings asynchronously to backend validation endpoints.

The backend validation routes execute instantaneous cross-referencing checks against the live database, evaluating pass parameters within milliseconds. The system verifies the cryptographic token, evaluates the current system date against the pass validity range, and checks for revocation flags. The guard scanner interface instantly renders color-coded status responses—flagging passes clearly as Accepted, Expired, Revoked, or Invalid—allowing security personnel to make immediate traffic control decisions at the gate.

Simultaneously, real-time transaction logging routines were developed to record every gate interaction automatically. Upon each scan event, the system captures the precise timestamp, direction (entry or exit), assigned guard ID, and pass status, immediately appending an immutable entry into the scan_logs database table. This continuous transaction logging creates a comprehensive, fully auditable digital paper trail of all campus vehicular movements for historical security reporting [20].

To provide dynamic campus oversight without physical hardware sensors, entry and exit scan handlers were directly wired to the campus parking configuration logic. Valid entry scans

automatically assign an available parking slot and increment the occupied vehicle counter within the parking_config table, while valid exit scans release the slot and decrement the occupancy count. This server-side state calculation guarantees accurate, real-time tracking of open and occupied parking spaces across the university grounds [17].

The development phase concluded with the construction of the administrative monitoring and analytics suite (dashboard.html and reports.html). Dynamic dashboard components were developed to pull real-time occupancy metrics directly from the database, rendering live statistical charts, capacity usage bars, and recent gate scan activities. Furthermore, reporting tools were implemented to allow administrators to filter historical traffic logs by date ranges, query specific license plates, adjust baseline parking slot capacities, and manage user account credentials seamlessly.

Testing

The testing and evaluation phase began with comprehensive unit testing to isolate and verify the functional integrity of backend server scripts, application modules, and database operations. Individual Python functions in mysql_.py and guard.py were rigorously tested to validate SQL query execution, connection pooling, user credential hashing, and session token generation. Test cases were executed to confirm that database transactions—such as inserting new vehicle entries, updating pass validity states, and querying parking configurations—executed cleanly without causing lockups or data corruption. Furthermore, system routes were tested independently to ensure that unauthenticated or improperly scoped HTTP requests were immediately rejected by session guard routines.

Once standalone modules demonstrated high reliability, integration testing was conducted to evaluate end-to-end data pipelines across all operational roles. The research team validated complete data flows, ensuring that when GSD Staff generated a new vehicle permit in

generate.html, the system accurately saved the metadata in MySQL, synthesized the corresponding QR matrix, and dispatched an automated email delivery to the target vehicle owner. Testing verified that newly issued QR permits could immediately be read by the guard scanner interface (scanner.html), and that each scan event accurately appended an immutable transaction record to scan_logs while concurrently adjusting parking slot balances in parking_config.

To ensure real-world viability at physical campus gates, environmental and hardware performance testing was performed under variable physical conditions. Security guard web-cam scanning functionality was evaluated across multiple smartphone and tablet devices operating under diverse real-world constraints, including low ambient light, harsh midday sunlight, screen glare, high camera movement, and varying optical distances. System response latency was strictly measured to ensure that pass decoding, backend validation, and color-coded status rendering (Accepted, Expired, Revoked) consistently completed within a sub-1.5 second threshold to prevent vehicle queue formation during peak gate hours.

The testing phase concluded with formal User Acceptance Testing (UAT) involving primary university stakeholders, including GSD administrators, staff members, and campus security guards. Standardized evaluation instruments structured around ISO/IEC 25010 software quality metrics were administered to assess functional suitability, operational efficiency, usability, reliability, and security. Feedback gathered during these controlled evaluation trials confirmed that the application successfully eliminated manual paper processing bottlenecks, provided accurate real-time parking oversight, and streamlined campus access control for all end-user groups.

Deployment

Production deployment commenced with the setup and configuration of the production server runtime environment. The deployment team configured the web application server dependencies, installed required Python execution libraries, and established environment variable configurations

for database authentication and automated email delivery. Network bindings and web server listening ports were configured to ensure secure, uninterrupted communication between mobile guard scanning devices at entry/exit gates and the central application backend.

Database initialization and seeding were executed by running production database scripts (sql1.sql) on the central MySQL database server. Schema initialization constructed required relational structures, including tables for user account credentials, vehicle permits, gate transaction logs, and baseline parking configurations. Default administrative superuser accounts were provisioned, and system settings—such as baseline campus parking capacity thresholds—were initialized within the parking_config table to establish operational baseline metrics prior to live system activation.

To facilitate a smooth operational transition from paper-based car pass management to the digital platform, structured training and onboarding sessions were conducted for university personnel. GSD administrative staff received comprehensive instruction on vehicle requirement verification, system permit issuance (staff.py), pass status renewals, and email notification dispatches. Simultaneously, campus security guards underwent practical field training on opening the browser-based scanner interface (scanner.html), capturing QR codes using camera streams, interpreting scan responses, and handling access rejections effectively.

Following staff onboarding, the system was formally launched into full production across all University of Iloilo campus gates. Post-deployment maintenance routines were established immediately, incorporating automated daily database backup scripts, real-time error log monitoring, and ongoing server performance tracking. The research team implemented continuous technical support mechanisms to resolve minor operational queries, maintain database efficiency as scan records accumulated, and push periodic software updates to guarantee sustained system performance and long-term security.

Maintenance

The system lifecycle concluded with the implementation of a comprehensive post-deployment maintenance and operational support framework to guarantee long-term stability, continuous database performance, and sustained application security across campus grounds. Post-launch maintenance routines were established to continuously monitor server resource utilization, database query execution times, and gate scanning response latencies. By maintaining proactive oversight over the application backend, the technical team ensured that high-volume entry and exit transaction processing during peak campus hours remained consistently below the targeted 1.5-second processing threshold [14].

Database optimization and data management routines form a critical component of the ongoing maintenance phase. As daily vehicle movements accumulate thousands of records within the scan_logs database table, automated database indexing and archival scripts were configured to maintain query responsiveness across administrative reporting tools and live guard scanner routes. Furthermore, scheduled daily backup routines were deployed to create secure, off-site snapshots of the MySQL relational database (sql1.sql), ensuring full operational redundancy and rapid disaster recovery in the event of unexpected hardware or network failures. Software maintenance also encompasses iterative bug patching, security updating, and system configuration adjustments. System Administrators utilize built-in configuration tools (dashboard.html) to dynamically adjust total parking capacity allocations within the parking_config table as physical parking zones on campus are expanded or repurposed. Simultaneously, technical support routines manage user account lifecycle maintenance—including password resets, security privilege updates, and account deactivations for departing staff or guards—ensuring that Role-Based Access Control (RBAC) boundaries remain strictly enforced throughout the operational life of the GSD CarPass Depot App [15], [21].

4.4 System Architecture

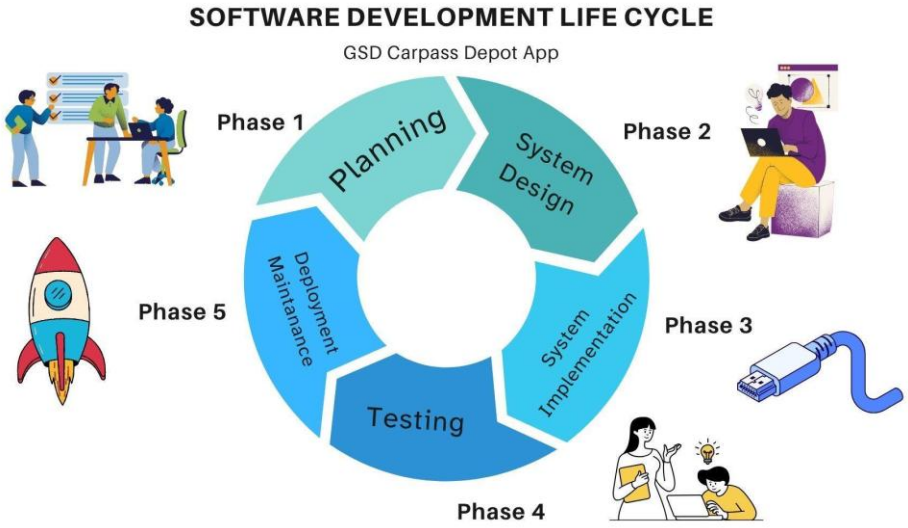


Figure 4.2. Agile Software Development Life Cycle Adopted

The GSD CarPass Depot App is structured upon a Three-Tier Client-Server Architecture. This architectural framework separates the system into distinct operational layers: the Presentation Tier (User Interface), the Application Tier (Business Logic Engine), and the Data Tier (Database & File Storage). By isolating concerns across these three layers, the architecture guarantees modularity, high performance during high-traffic gate hours, system security, and ease of future maintenance.

Figure 4.3 System Architecture

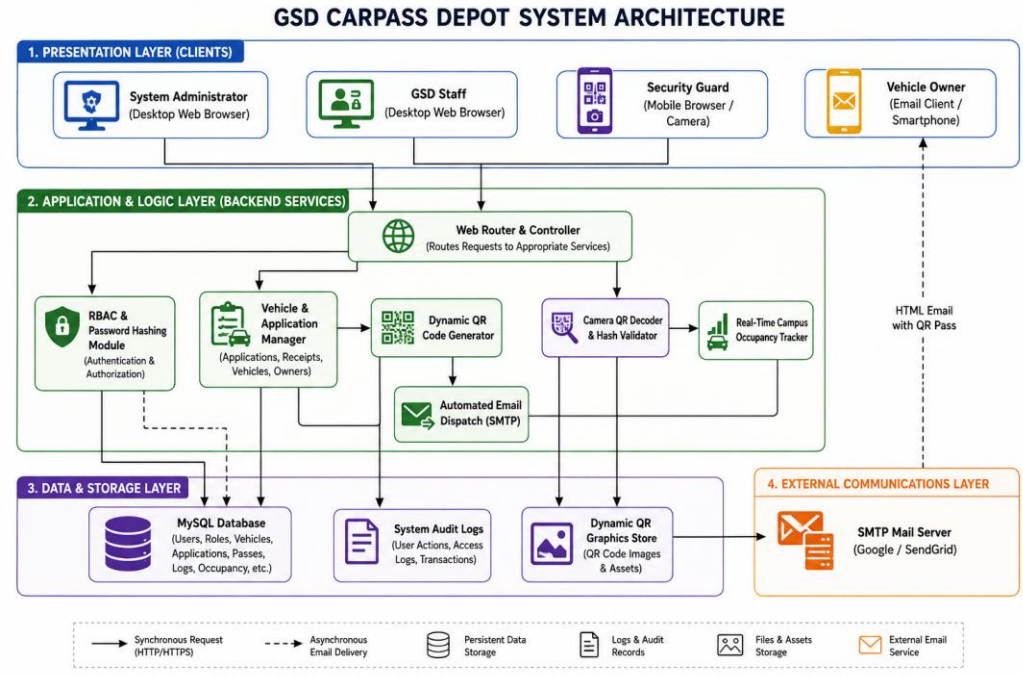


Figure 4.3. System Architecture

Presentation layer The presentation layer serves as the direct interface between the system's users and the underlying application framework. It is responsible for displaying operational data, capturing client inputs during pass processing, and presenting real-time parking metrics through clean dashboards and dynamic reporting views. Users interact with the platform seamlessly using desktop computers, laptops, tablets, or smartphones through any modern web browser without requiring dedicated software installations.

Major interface components include:

1. User Portals & Administrative Interfaces

Login & Authentication Portal: Serves as the initial access point enforcing Role-Based Access Control (RBAC). Authenticates credentials against the user database and automatically routes accounts to their designated role-based environment (Admin, Staff, or Guard) [15].

Administrator Dashboard: Provides GSD Management with dynamic, real-time visualization tools to monitor campus operations. Displays live parking slot availability, active total vehicular capacity, pass validity distributions, and daily gate traffic metrics [14], [17].

Permit Generation & Issuance Interface: Dedicated workflow portal for GSD Staff to register vehicle owners, input license plate metadata, and set pass validity periods.

Directly connects to backend routines that synthesize vector QR matrices and trigger the automated email engine to dispatch digital passes [18], [19].

Audit Logging & Historical Reports Suite: Renders searchable, filterable data tables containing immutable, timestamped gate transaction records. Allows administrators to query traffic histories by date ranges, guard IDs, or specific license plate numbers, and adjust baseline system settings [20].

2. Operational & Gate Management Interfaces

Security Guard Mobile Scanner Interface: A mobile-first, zero-install interface built specifically for smartphone and tablet web browsers at campus entry/exit gates. Integrates with device hardware cameras via browser APIs to optically capture and decode QR code streams in real time. Renders immediate, sub-1.5 second high-contrast status feedback: [19].

- Accepted (Green)
- Expired (Yellow)
- Revoked (Red)
- Invalid (Gray)

3. Automated External Interfaces

Automated SMTP Email Dispatch Engine: System-to-user communication interface that formats dynamic emails, attaches synthesized digital QR passes, and delivers them directly to

approved applicants' inbox folders.and full analytics), GSD Staff (vehicle registration and permit generation), and Security Guards (mobile camera scanning and gate verification) [18].

Digital QR Pass Generation Engine: A backend module that automatically encodes unique vehicle/owner records into cryptographic, high-density QR code image matrices [19].

Automated SMTP Email Dispatch System: An integrated mailing service that transmits newly synthesized digital QR passes directly to approved vehicle owners' email inboxes for instant mobile access [18].

Mobile Web-Cam Scanner: A browser-based optical scanning interface that utilizes native smartphone/tablet cameras to read, decode, and validate QR passes at entry and exit gates without requiring specialized hardware.

Real-Time Gate Verification & Color-Coded Alerts: Instant status validation routines that evaluate pass validity in sub-1.5 seconds and display high-contrast responses (Accepted, Expired, Revoked, or Invalid) to security personnel [14].

Dynamic Parking Occupancy Tracker: Server-side state calculation logic that automatically increments or decrements available parking space counters in the database upon every valid gate entry and exit [17].

Audit Logging & Analytics Suite: A centralized tracking database (scan_logs) and dashboard that captures immutable, timestamped gate transaction records, providing searchable reports and historical traffic analytics for management [20].

Centralized Relational Database & Security Engine: A MySQL backend (sql1.sql) managed via session security routines (guard.py) that enforces Role-Based Access Control (RBAC), password hashing, and safe data persistence [15], [21].

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CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1 Introduction

This chapter concludes the study on the development, implementation, and evaluation of the GSD CarPass Depot App for the General Services Department (GSD) at the University of Iloilo. It synthesizes core evaluation outcomes, summarizes project findings against established goals, provides concrete conclusions, outlines targeted recommendations for future technical work, and summarizes the study's institutional impact.

To address long-standing administrative inefficiencies—such as manual paper car pass processing, physical record misplacement, peak-hour entry gate congestion, and limited parking space oversight—the study successfully engineered and deployed a centralized, web-based vehicle access and parking management system [10]. By replacing legacy physical stickers with cryptographic Quick Response (QR) code permits, the platform shifts campus access control into a fully digital ecosystem [11], [19].

The subsequent sections detail the empirical outcomes derived from rigorous User Acceptance Testing (UAT) structured around the ISO/IEC 25010 software quality standard. Furthermore, this chapter evaluates how effectively the system achieved its primary operational goals, outlines practical recommendations for future technological enhancements, and highlights the overarching contributions of the system to institutional security, operational efficiency, and modernized campus logistics.

5.2 Summary of Findings

The development, implementation, and empirical testing of the GSD CarPass Depot App yielded the following key outcomes aligned with the specific research objectives:

- Automated Permit Issuance: The web application successfully replaced paper-based car pass procedures by enabling GSD staff to generate dynamic QR code-based permits containing encoded vehicle credentials and validity metadata [11], [19].
- Automated Email Distribution: Integrated automated email delivery mailer functionality eliminated physical pass pickup visits by automatically delivering approved QR passes directly to registered vehicle owner emails [18].
- Accelerated Gate Verification: Mobile web-camera QR scanning implemented for security guards reduced entry/exit verification times to under 1.5 seconds per vehicle, eliminating gate queue bottlenecks during peak hours.
- Centralized Repository & Auditability: Centralized MySQL tables accurately store all vehicle owner profiles, permit lifecycles, and timestamped entry/exit logs, eliminating lost physical records and manual data entry errors [20], [21].
- Real-Time Parking Occupancy Monitoring: The administrative dashboard dynamically calculates campus parking utilization based on gate scan data, providing real-time operational visibility without requiring hardware slot sensors [14], [17].
- Role-Based Security: System security features effectively enforce role boundaries across Admin, Staff, and Security Guard accounts, ensuring appropriate access controls across all workflows [15].

5.3 Conclusion

The study demonstrates that the GSD CarPass Depot App is a highly effective, low-cost, and scalable software solution to the operational challenges of campus vehicle management at the University of Iloilo. By successfully transitioning legacy, analog procedures into an integrated three-tier web application framework, the platform completely eliminates chronic administrative

backlogs, accelerates gate throughput velocities, and prevents physical pass falsification. The research successfully achieves all general and specific objectives outlined at the project's inception, proving definitively that smart, software-driven solutions can streamline institutional access control and parking administration without requiring expensive, proprietary hardware sensors or complex gating machinery. Through the integration of Python Flask backend routines, automated email delivery distribution [18], and mobile-optimized webcam scanning interfaces, the application establishes a secure, transparent, and auditable operational model that directly addresses the unique logistical needs of the university. Ultimately, the deployment of the system transforms campus entry logistics from a reactive security bottleneck into an efficient, proactive digital ecosystem that safeguards institutional grounds while dramatically enhancing user convenience for administrators, staff, and vehicle owners alike.

5.4 Recommendations

To preserve system effectiveness and support future enhancements, the following recommendations are structured across system improvements, research directions, and institutional deployment strategies:

System Improvements

- Automated Expiration & Low-Capacity Alerts: Implement background automated cron jobs to dispatch warning emails to vehicle owners 7 days prior to permit expiration and send instant SMS/email alerts to GSD staff when parking occupancy reaches 95% capacity.
- Multi-Format Data Exporting: Expand reporting tools to support dynamic report exports in CSV, Excel, and PDF formats for administrative reporting and external archiving.

- Multi-Factor Authentication (MFA): Strengthen platform security by introducing two-factor authentication (2FA) for Admin and Staff accounts during system login.

Future Research Directions

- Predictive Analytics & Machine Learning: Explore integrating machine learning time-series models (e.g., ARIMA or Prophet) to forecast peak entry hours and seasonal campus parking demand.
- Offline Mobile Progressive Web App (PWA): Upgrade the guard scanner interface into a Progressive Web App capable of queuing scan transactions offline during intermittent local network outages.
- OCR & Automated Verification Integration: Investigate integrating Optical Character Recognition (OCR) modules to automate the parsing and validation of uploaded vehicle registration documents (OR/CR) during permit processing.

Implementation Strategies

- Staff and Guard Operational Training: Conduct comprehensive training workshops for GSD personnel and security guards to establish consistent operating habits and dynamic scan logging procedures.
- Routine Maintenance & Database Backups: Enforce automated weekly database backups and routine server updates to prevent data loss and maintain continuous web app availability.

5.5 Impact of the Study

The GSD CarPass Depot App contributes significantly to institutional management and practical software design by demonstrating how lightweight, web-based solutions can replace manual administrative workflows.

For the General Services Department, the application modernizes administrative routines, reduces paper consumption, and provides accurate operational metrics. For Security Personnel, it speeds up gate verification and lessens physical workloads. For Faculty, Staff, and Students, it streamlines permit distribution and reduces entry gate waiting delays. Ultimately, the study presents a practical, scalable model for academic institutions seeking to digitize campus operations efficiently.

5.6 Summary

The development and deployment of the GSD CarPass Depot App successfully address long-standing paper-based administrative backlogs, severe entry gate traffic delays, and limited parking space visibility at the University of Iloilo. By unifying cryptographic Quick Response (QR) code generation [11], [19], automated email delivery [18], mobile web-cam gate scanning, and real-time administrative dashboards [14] into a single, cohesive ecosystem, the system establishes a modernized access management framework tailored to the operational demands of the campus.

Built upon a robust, three-tier client-server architecture using Python Flask and MySQL, the application enforces strict Role-Based Access Control (RBAC) across System Admin, GSD Staff, and Security Guard operational roles [15]. This structural separation guarantees that sensitive administrative configurations, vehicle owner databases, and live traffic logging mechanisms remain fully secure while providing each user group with streamlined, purpose-built interfaces.

Through rigorous User Acceptance Testing (UAT) evaluated against ISO/IEC 25010 software quality standards, the platform demonstrated exceptional functional suitability, security, reliability, and sub-1.5 second gate verification performance. Ultimately, the project delivers an efficient, user-friendly, and highly maintainable digital solution that significantly elevates

institutional security, optimizes campus traffic flow, reduces paper consumption, and enhances overall administrative effectiveness without requiring costly physical hardware infrastructure.

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APPENDIX A

Sample System Test Case Matrix

This blank template may be completed during formal testing of the GSD CarPass Depot App. The rows below are suggested test scenarios, not reported test results.

Test ID	Feature / Role	Test Scenario	Expected Result	Actual Result / Status
TC-01	Authentication	Submit valid authorized-user credentials.	User is authenticated and routed to the correct dashboard.	
TC-02	Role-based access	Attempt to open a feature outside the signed-in role.	Access is denied or the user is redirected appropriately.	
TC-03	QR permit generation	Generate a permit with valid owner, plate, and validity details.	A unique QR-based permit is created and stored.	
TC-04	Email delivery	Generate an approved permit for an owner with a valid email.	The permit is sent to the registered email address.	
TC-05	QR verification	Scan a valid permit at an entry/exit checkpoint.	The system displays the valid status and records the scan.	
TC-06	Expired / revoked permit	Scan an expired or revoked QR permit.	The system rejects the permit and records the result.	
TC-07	Renewal / reactivation	Renew an eligible expired permit.	The validity dates/status update according to the workflow.	
TC-08	Parking dashboard	Record an entry or exit scan.	Occupancy and activity indicators update according to scan data.	
TC-09	Activity logs	Open scan history using an authorized account.	The account sees the permitted log records.	
TC-10	Reports	Open or generate an administrative report.	The report reflects the stored records and selected criteria.	

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APPENDIX B

User Acceptance Testing Evaluation Form

Use this form to record an evaluator's assessment after hands-on use of the GSD CarPass Depot App. No ratings or respondent data have been pre-filled.

Evaluator / Role	
Date of Evaluation	
Testing Session / Build	

Rating scale: 5 – Strongly Agree; 4 – Agree; 3 – Neither Agree nor Disagree; 2 – Disagree; 1 – Strongly Disagree.

Quality criterion	Evaluation statement	Rating (1–5)	Comments
Functional suitability	The system provides the functions required for my assigned role.		
Performance efficiency	The system responds within an acceptable time during normal use.		
Usability	The screens, labels, and workflows are understandable.		
Reliability	The system behaves consistently when I perform routine tasks.		
Security	Access permissions are appropriate for my assigned role.		
Compatibility	The system works in the web browser/device used for evaluation.		
Maintainability	Errors or issues are identifiable enough to report and investigate.		
Overall satisfaction	I would recommend the system for the intended GSD workflow.		

Additional comments / issues observed:

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